

# Differentially Private Federated Best Arm Identification

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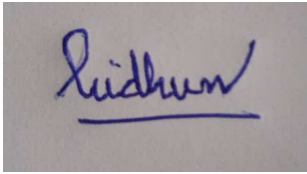
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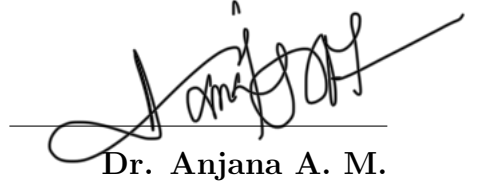
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## DEDICATION

*I dedicate my thesis to my beloved Parents.*

# ABSTRACT

This thesis studies the problem of Federated, Differentially Private Best Arm Identification (DP-BAI). We give an information-theoretic lower bound on the sample complexity of the problem, present a communication efficient federated algorithm and analyze its performance through simulations. This study highlights the interplay between privacy, communication, and statistical efficiency in federated decision-making.

**Keywords:** Multi-armed bandits, Best arm identification, Differential privacy

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# Chapter 1

## Introduction

**The** study of active sequential hypothesis testing can be traced back to the pioneering work of Abraham Wald and Herman Chernoff in the mid-twentieth century. Their contributions established the foundations of sequential decision-making, where observations are collected adaptively and decisions are made as soon as sufficient evidence becomes available. Over the decades, these ideas have evolved into a broad research area spanning statistics, information theory, operations research, and machine learning.

The multi-armed bandit framework is central to studying sequential decision-making and the exploration-exploitation trade-off. Originally designed for resource allocation, it now underpins modern reinforcement learning and online learning, with applications ranging from clinical trials and recommender systems to adaptive experiments.

Within this domain, Best Arm Identification (BAI) stands out as a core pure-exploration task. BAI methodologies rely on sequential hypothesis testing, information-theoretic lower bounds, and adaptive sampling. To set the stage for the specific problem addressed in this thesis, this section reviews the foundational concepts, formulations, and theoretical results of the classical BAI problem.

### 1.1 Multi Armed Bandits

A multi-armed bandit model is a canonical framework for studying sequential decision-making under uncertainty. The model consists of  $K$  probability distributions  $\nu_1, \dots, \nu_K$ , referred to as arms, with corresponding mean rewards  $\mu_1, \dots, \mu_K$ . At each time step  $t = 1, 2, \dots$ , the learner selects an arm  $A_t \in \mathcal{A} = \{1, \dots, K\}$  and receives a reward

$X_t$  sampled independently from the distribution  $\nu_{A_t}$  [1]. Since the underlying distributions are unknown, the learner must base its decisions solely on the history of previously selected arms and observed rewards.

The central challenge is to design a sampling strategy that balances information acquisition and performance in order to achieve a prescribed objective. Broadly speaking, objectives in multi-armed bandit problems fall into two categories. In regret minimization, the learner seeks to maximize the cumulative reward collected over time, or equivalently minimize the loss incurred by not always selecting the optimal arm. In contrast, best arm identification focuses on accurately determining the arm with the highest mean reward while using as few samples as possible. The latter setting, often referred to as a pure-exploration problem, forms the primary focus of this thesis.

## 1.2 Best Arm Identification

The objective is to identify the arm with the largest mean reward, denoted by  $\mu^* = \max_{1 \leq a \leq K} \mu_a$ , as quickly and accurately as possible. Unlike regret minimization, the quality of a strategy is evaluated solely through its ability to correctly identify the optimal arm, regardless of how many times suboptimal arms are sampled during the exploration phase.

Let  $F_t = \sigma(X_1, \dots, X_t)$  denote the sigma-field generated by the observations collected up to time  $t$ . A best arm identification strategy is characterized by three components:

- A sampling rule  $(A_t)_t$ , where  $A_t$  is  $F_{t-1}$ -measurable;
- A stopping rule  $\tau$ , which is a stopping time with respect to the filtration  $(F_t)_t$ ;
- A decision rule  $\hat{a}_\tau$ , which is  $F_\tau$ -measurable.

The objective is to maximize the probability that the recommended arm  $\hat{a}_\tau$  belongs to the set of optimal arms,  $\hat{a}_\tau \in \arg \max_{a \in \mathcal{A}} \mu_a$ , while minimizing the number of samples  $\tau$  required before making a decision [1].

Two principal settings have been extensively studied in the literature. In the *fixed-budget* setting, the total number of samples is specified in advance, and the goal is to minimize the probability of error  $\mathbb{P}(\hat{a}_\tau \notin \arg \max_{a \in \mathcal{A}} \mu_a)$  [1].

In the *fixed-confidence* setting, a confidence level  $\delta \in (0, 1)$  is prescribed beforehand, and one seeks a strategy satisfying  $\mathbb{P}(\hat{a}_\tau \notin \arg \max_{a \in \mathcal{A}} \mu_a) \leq \delta$  [1].

Such a strategy is said to be  $\delta$ -PAC (Probably Approximately Correct). Among all  $\delta$ -PAC strategies, the objective is to minimize the expected sample complexity  $\mathbb{E}[\tau]$  [1]. Our focus is on the best arm identification with fixed confidence setting.

### 1.3 Lower bound for the vanilla BAI problem

The study of best arm identification begins with the characterization of the instance-dependent lower bound on sample complexity. Beyond establishing a fundamental performance limit that no algorithm can surpass, this lower bound provides valuable insights into the structure of optimal sampling strategies. In particular, it plays a crucial role in the design of asymptotically optimal algorithms, guiding both the allocation of samples among arms and the construction of statistically sound stopping rules.

Denote by  $\mathcal{S}$  a set of exponential bandit models. Let  $\boldsymbol{\mu} = (\mu_1, \dots, \mu_K)$  denote a bandit instance. We assume that there exists a unique optimal arm, denoted by  $a^*(\boldsymbol{\mu}) = \arg \max_{i \in [K]} \mu_i$ . Fix a confidence level  $\delta \in (0, 1)$ . Let  $d(\mu, \mu')$  denote the Kullback-Leibler divergence between two distributions parametrized by  $\mu$  and  $\mu'$ . Then, for any  $\delta$ -PAC strategy, the expected stopping time satisfies

$$\mathbb{E}_{\boldsymbol{\mu}}[\tau_\delta] \geq T^*(\boldsymbol{\mu}) \text{kl}(\delta, 1 - \delta),$$

where the characteristic time  $T^*(\boldsymbol{\mu})$  is defined through

$$\frac{1}{T^*(\boldsymbol{\mu})} = \sup_{w \in \Sigma_K} \inf_{\boldsymbol{\lambda} \in \text{Alt}(\boldsymbol{\mu})} \left( \sum_{a=1}^K w_a d(\mu_a, \lambda_a) \right),$$

and

$$\text{Alt}(\boldsymbol{\mu}) = \{\boldsymbol{\lambda} \in \mathcal{S} : a^*(\boldsymbol{\lambda}) \neq a^*(\boldsymbol{\mu})\}.$$

The set  $\text{Alt}(\boldsymbol{\mu})$  consists of all alternative bandit instances whose optimal arm differs from that of the true instance and  $w$  is a probability distribution over the arms. The quantity  $T^*(\boldsymbol{\mu})$  measures the intrinsic difficulty of distinguishing the true instance from its closest alternatives. Consequently, it determines the minimum number of

samples required by any algorithm to identify the optimal arm with confidence at least  $1 - \delta$  [1].

The expression defining  $T^*(\boldsymbol{\mu})$  admits a natural game-theoretic interpretation. Consider a two-player zero-sum game between the learner and nature. The learner selects a sampling proportion vector  $w \in \Sigma_K$ , which determines how observations are allocated among the arms. In response, nature chooses an alternative instance  $\boldsymbol{\lambda} \in \text{Alt}(\boldsymbol{\mu})$  that is hardest to distinguish from the true instance. For a fixed allocation  $w$ , the quantity  $\sum_{a=1}^K w_a d(\mu_a, \lambda_a)$  represents the average information gathered per sample for discriminating between  $\boldsymbol{\mu}$  and  $\boldsymbol{\lambda}$ . Nature seeks to minimize this information, whereas the learner seeks an allocation that maximizes the resulting value. The sup–inf structure therefore captures the largest discrimination rate that can be guaranteed against the most challenging alternative instance. The optimal allocation  $w^*(\boldsymbol{\mu})$  emerging from this game determines the asymptotically optimal proportion of samples that should be allocated to each arm.

The analysis and proof of the above result is discussed in detail in [1]. Especially, the change of measure technique which allows to prove non-asymptotic lower bound on the sample complexity is of great importance. The information processing inequality from [1] given below would be used later in the discussion of our specific problem setting.

$$\forall \boldsymbol{\lambda} \in \mathcal{S} : a^*(\boldsymbol{\lambda}) \neq a^*(\boldsymbol{\mu}) , \quad \sum_{a=1}^K d(\mu_a, \lambda_a) \mathbb{E}_{\mu}[N_a(\tau_{\delta})] \geq \text{kl}(\delta, 1 - \delta).$$

Further simplification of the lower bound depending on the problem setup is possible. We would do that in detail for our problem later.

## 1.4 Important BAI algorithms

Two broad algorithmic paradigms dominate the fixed-confidence best arm identification literature. The first consists of tracking-based methods, introduced by Garivier and Kaufmann in [1], which aim to match the asymptotically optimal sampling proportions prescribed by the lower bound. The second consists of Top-Two algorithms, proposed by Russo in [2], which rely on maintaining and comparing a leading candidate arm against a carefully selected challenger. Both approaches achieve asymp-

totically optimal performance, but they differ substantially in the way samples are allocated and decisions are made.

### 1.4.1 Track and Stop

Track-and-Stop algorithms are built upon the observation that the lower bound not only characterizes the minimum achievable sample complexity but also prescribes how samples should be allocated among the arms. The optimal allocation vector  $w^*(\boldsymbol{\mu})$  identifies the proportions in which the arms should be sampled in order to gather evidence against the most difficult alternative instances as efficiently as possible. An asymptotically optimal algorithm should employ a sampling rule that tracks these proportions and a stopping rule that terminates as soon as the accumulated evidence is sufficient to certify the identity of the best arm.

The C-Tracking and D-Tracking algorithms from [1] are built directly on this idea. Because the true bandit instance  $\boldsymbol{\mu}$  is unknown, a learner cannot simply calculate the optimal allocation  $w^*(\boldsymbol{\mu})$ . Instead, the algorithm uses empirical means at each round to estimate the underlying instance as  $\hat{\boldsymbol{\mu}}(t)$ . Solving the characteristic time optimization problem with this estimate yields  $w^*(\hat{\boldsymbol{\mu}}(t))$ . The algorithm then makes sampling decisions to force the actual proportions of arm draws to track these estimated optimal targets. The analysis in [1] works largely because the optimization problem defining  $T^*(\boldsymbol{\mu})$  is concave with respect to the allocation  $w$ , and the resulting optimizer  $w^*(\boldsymbol{\mu})$  is continuous with respect to the bandit instance. Because the empirical mean estimates are consistent, these properties guarantee that  $w^*(\hat{\boldsymbol{\mu}}(t))$  converges to  $w^*(\boldsymbol{\mu})$  almost surely. Consequently, both tracking methods eventually allocate samples matching the lower-bound proportions, securing their asymptotic optimality. Yet, despite these excellent asymptotic guarantees, Track-and-Stop algorithms can be difficult to deploy in practice. The main bottleneck is having to resolve the lower-bound optimization problem at every single round just to find  $w^*(\hat{\boldsymbol{\mu}}(t))$ . For most bandit models, this optimization lacks a closed-form solution. Solving it numerically at each step quickly becomes computationally expensive, especially as the number of arms scales up.

A second issue is that tracking-based methods rely on maintaining the entire allocation vector prescribed by the lower bound. While this is theoretically appealing, the resulting sampling rule can be difficult to interpret and implement. Moreover, the optimization defining  $w^*(\boldsymbol{\mu})$  is global in nature, requiring information from all arms even when only a few arms remain plausible candidates for being optimal.

### 1.4.2 Top-Two Sampling

Top-Two Sampling frames best arm identification as a battle between a leader and a challenger. At each round, the algorithm flags a leader arm (the current favorite for being the best) and pairs it with a challenger arm meant to test that hypothesis. Usually, the leader is picked via an upper confidence bound (UCB) criterion, while the challenger is chosen using a transportation-cost (TC) principle to find whichever arm yields the most informative comparison. The algorithm then tosses a biased coin, sampling the leader with probability  $\beta \in (0, 1)$  and the challenger with probability  $1 - \beta$ . By constantly narrowing its focus to this pair, the method quickly accumulates data to either validate the leader or elevate a new contender. The real draw of Top-Two Sampling is how straightforward it is. It completely skips the heavy computational lifting of tracking-based methods, which force you to solve a complex lower-bound optimization problem at every single step. Instead, you just identify the leader and challenger, keeping the entire process lightweight and easy to code. By isolating exploration to just a tiny subset of statistically relevant arms, Top-Two methods generally perform incredibly well in practice and easily scale to problems with a huge action space.

The sampling rule of Top-Two algorithms does not explicitly track the optimal allocation vector  $w^*(\mu)$  prescribed by the lower-bound analysis. Instead, the long-run allocation is determined by the choice of the tuning parameter  $\beta$  and the interaction between the leader and challenger dynamics. As a result, the asymptotic sampling proportions generally differ from the information-theoretically optimal allocation.

This discrepancy has important consequences for performance guarantees. While Top-Two algorithms achieve the correct logarithmic scaling of the sample complexity with respect to the confidence parameter, their characteristic time may differ from the optimal value by a multiplicative constant. In other words, they are asymptotically optimal only up to a constant factor. The simplicity and low computational cost of Top-Two methods therefore come at the expense of not fully exploiting the optimal allocation structure revealed by the lower-bound analysis[2].

## 1.5 Stopping rule

The stopping rule in Track-and-Stop is closely tied to the information-theoretic lower bound discussed earlier. Recall that the lower bound is obtained by considering the



amount of information required to distinguish the true instance from every alternative instance with a different optimal arm. This suggests that an algorithm should stop only when the observations collected so far provide sufficient evidence against all such alternatives.

To formalize this idea, Garivier and Kaufmann [1] introduce a generalized likelihood ratio (GLR) statistic. At time  $t$ , let  $\hat{\boldsymbol{\mu}}(t)$  denote the vector of empirical means and let  $\hat{a}(t)$  be the empirical best arm. The statistic measures the evidence in favor of the hypothesis that  $\hat{a}(t)$  is indeed the optimal arm against the alternative hypothesis that some other arm is optimal. For one-parameter exponential families, this statistic can be written as

$$Z(t) = \min_{b \neq \hat{a}(t)} \left\{ N_{\hat{a}(t)}(t) d(\hat{\boldsymbol{\mu}}_{\hat{a}(t)}(t), \hat{\boldsymbol{\mu}}_{\hat{a}(t),b}(t)) + N_b(t) d(\hat{\boldsymbol{\mu}}_b(t), \hat{\boldsymbol{\mu}}_{\hat{a}(t),b}(t)) \right\},$$

where  $N_a(t)$  denotes the number of samples collected from arm  $a$  up to time  $t$  and  $\hat{\boldsymbol{\mu}}_{\hat{a}(t),b}(t)$  is the weighted average of the empirical means of arms  $\hat{a}(t)$  and  $b$ . The form of this statistic closely resembles the lower-bound expression, with the optimal proportions replaced by the actual numbers of pulls accumulated by the algorithm.

The stopping rule is then defined by comparing the GLR statistic against a threshold. Specifically, the algorithm stops at

$$\tau_\delta = \inf \{t \geq 1 : Z(t) > \beta(t, \delta)\},$$

where  $\beta(t, \delta)$  is an exploration threshold that depends on the desired confidence level  $\delta$ . The threshold is chosen so that the probability of recommending an incorrect arm is at most  $\delta$ . In other words, the stopping rule guarantees that the resulting algorithm is  $\delta$ -PAC.

The rationale behind this construction is analogous to classical sequential hypothesis testing. The quantity  $Z(t)$  measures the cumulative evidence gathered in favor of the current empirical best arm, while the threshold  $\beta(t, \delta)$  specifies how much evidence is required before a decision can be trusted. As additional samples are collected, the GLR statistic increases whenever the observations consistently support a particular arm. Once the evidence exceeds the prescribed threshold, the algorithm terminates and recommends the empirical best arm.

## 1.6 Sample Complexity analysis

The central objective in the analysis of fixed-confidence best arm identification algorithms is to characterize their sample complexity, that is, the expected number of samples required to identify the optimal arm with confidence at least  $1 - \delta$ . For asymptotically efficient algorithms, the sample complexity is expected to scale logarithmically with the confidence parameter, taking the form

$$\mathbb{E}_{\boldsymbol{\mu}}[\tau_{\delta}] \approx C(\boldsymbol{\mu}) \log \frac{1}{\delta},$$

where the instance-dependent constant  $C(\boldsymbol{\mu})$  quantifies the statistical difficulty of the problem. The lower bound discussed earlier establishes that no  $\delta$ -PAC algorithm can achieve a leading constant smaller than  $T^*(\boldsymbol{\mu})$ . Consequently, the primary goal of algorithm design and analysis is to attain this optimal constant or come as close to it as possible.

For Track-and-Stop, the sample complexity analysis relies on two key properties. First, the tracking rule guarantees that the empirical sampling proportions converge to the optimal allocation  $w^*(\boldsymbol{\mu})$  prescribed by the lower-bound optimization problem. Second, the generalized likelihood ratio stopping rule ensures that the algorithm terminates as soon as sufficient evidence has been accumulated to distinguish the true instance from all alternative instances. Combining these ingredients, Garivier and Kaufmann [1] prove that

$$\limsup_{\delta \rightarrow 0} \frac{\mathbb{E}_{\boldsymbol{\mu}}[\tau_{\delta}]}{\log(1/\delta)} \leq T^*(\boldsymbol{\mu}),$$

which matches the information-theoretic lower bound. Thus, Track-and-Stop is asymptotically optimal, achieving the smallest possible leading constant in the sample complexity.

The analysis of Top-Two Sampling proceeds differently. Since the algorithm does not explicitly track the optimal allocation, its asymptotic behavior is governed by the limiting allocation induced by the leader–challenger dynamics and the parameter  $\beta$ . Under suitable conditions, the empirical sampling proportions converge to a deterministic allocation, yielding an information acquisition rate that determines the stopping time. Russo [2] shows that the expected sample complexity also scales logarithmically with  $1/\delta$ , namely,

$$\mathbb{E}_{\boldsymbol{\mu}}[\tau_{\delta}] = O\left(\Gamma_{\beta}^*(\boldsymbol{\mu})^{-1} \log \frac{1}{\delta}\right),$$

where  $\Gamma_{\beta}^*(\boldsymbol{\mu})$  denotes the asymptotic rate at which evidence is accumulated. Although this rate generally differs from the optimal rate  $T^*(\boldsymbol{\mu})^{-1}$ , it remains within a constant factor of the optimum for an appropriate choice of  $\beta$ . Consequently, Top-Two Sampling preserves the optimal logarithmic dependence on the confidence parameter while trading a constant factor in sample complexity for a significantly simpler sampling rule.

This chapter introduced the fundamental concepts underlying fixed-confidence best arm identification. We discussed the multi-armed bandit framework, the information-theoretic lower bound governing sample complexity, and the algorithmic principles that have emerged from its analysis. We examined how the lower bound not only characterizes the intrinsic difficulty of an instance but also guides the design of sampling and stopping rules. We then reviewed two of the most influential approaches in the literature, namely Track-and-Stop and Top-Two Sampling, highlighting the trade-offs between statistical optimality and computational simplicity.

The developments presented in this chapter largely concern the classical centralized setting, where a single learner sequentially interacts with all arms and has unrestricted access to the collected observations. The next chapter introduces the distributed best arm identification problem that forms the focus of this thesis and explores how the principles developed for the classical setting can be adapted to this more challenging environment.

# Chapter 2

## Problem Setup

The classical best arm identification framework assumes that all observations are collected by a single learner and that the underlying reward distribution of each arm is identical across all samples. In many real-world applications data are generated by multiple agents or clients, each possessing its own local observations and evaluation criteria. The quality of an alternative may vary across clients due to differences in behavior, context, preferences, or measurement practices. Therefore, local estimates of arm quality may not accurately reflect the global objective, making it necessary to aggregate information across multiple clients.

One setting where such considerations arise is in market surveys. Consider a collection of retailers evaluating different product brands. Here, retailers correspond to clients and brands correspond to arms. Customer ratings observed by a particular retailer may be influenced by local advertising campaigns, demographic factors, purchasing habits, or subjective interpretations of product quality. As a result, a brand that appears superior from the perspective of one retailer may not necessarily be the best choice globally. Aggregating information across retailers provides a more reliable estimate of the overall quality of each brand and reduces the influence of localized biases.

A similar phenomenon appears in peer-review systems and democratic decision-making processes. In a conference review process, different reviewers may apply varying standards when evaluating submissions. Likewise, in elections, different regions or states may exhibit systematic preferences that do not necessarily reflect the broader population. Local decisions can therefore be distorted by favoritism, calibration differences, or regional effects. Combining information from multiple reviewers or regions helps mitigate these distortions and produces decisions that better represent

the collective assessment. A major hurdle here is that data from individual clients is often messy, biased, and inconsistent, even though our ultimate goal is to find a global metric by combining all this population-wide information.

On top of that, working in a distributed setup means we have to protect user privacy. Individual clients often handle sensitive data that can't just be sent over to a central server. For example, things like survey preferences, peer reviews, or personal recommendation data are often legally protected or highly proprietary. Any distributed "best arm identification" algorithm has to walk a fine line between statistical accuracy and strict privacy. This is exactly why federated learning has become so vital—it lets clients team up to solve a problem without exposing their raw data during communication.

This thesis sits right at the intersection of these two challenges: finding the globally optimal arm in a distributed network while strictly safeguarding each client's private information.

## 2.1 Setup

We consider a federated multi-armed bandit framework involving a central server and  $M$  clients. Each client interacts with an independent instance of the multi-armed bandit having  $K$  arms. We write  $[K] := \{1, 2, \dots, K\}$  to denote set of arms. We assume that the rewards generated from each arm of any given client are independent and identically distributed (i.i.d.), and independent of the rewards generated from other arms of the client and the arms of other clients. For each client  $m \in [M]$  and arm  $a \in [K]$ , the rewards are drawn from an unknown distribution  $\nu_{a,m}$  with mean  $\mu_{a,m} \in \mathbb{R}$ . A *problem instance*  $\nu = \{\nu_{a,m} : a \in [K], m \in [M]\}$  is specified by the collection of arm distributions. For each tuple  $(a, m)$ , we assume that the reward distribution  $\nu_{a,m}$  of arm  $a$  at client  $m$  belongs to the exponential family of probability distributions parametrised by its mean  $\mu_{a,m}$ . Throughout the paper, we keep the underlying instance  $\nu$  fixed.

At time step  $t \geq 1$ , client  $m$  selects an arm  $A_{t,m}$  and observes a reward  $R_{t,m} \sim \nu_{A_{t,m},m}$ .

In this thesis, we consider two notions of best arm, as defined below.

1. **Local Best Arm:** The *local best arm* of client  $m$  under the instance  $\nu$ , denoted by  $a_m^*(\nu)$ , is defined as the arm with the highest mean in the bandit instance

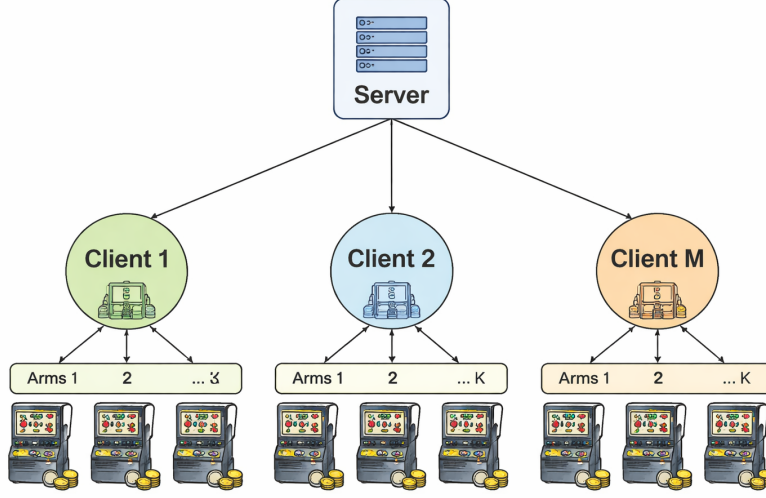


Figure 2.1: Illustration of the federated best arm identification framework. Multiple clients interact with their local environments and communicate with a central server to identify the globally optimal arm.

corresponding to client  $m$ , i.e.,

$$a_m^*(\nu) := \arg \max_{a \in [K]} \mu_{a,m}. \quad (2.1)$$

Without loss of generality, we assume that the local best arm of each client is uniquely specified.

2. **Global Best Arm:** The *global best arm* under the instance  $\nu$ , denoted by  $a^*(\nu)$ , is defined as

$$a^*(\nu) := \arg \max_{a \in [K]} \mu_a, \quad (2.2)$$

where

$$\mu_a := \frac{1}{M} \sum_{m=1}^M \mu_{a,m}. \quad (2.3)$$

Without loss of generality, we assume that the global best arm is uniquely specified.

We write  $\mathbf{a}^*(\nu) = (a_1^*(\nu), \dots, a_M^*(\nu), a^*(\nu))$  to denote the tuple of best arms under the

problem instance  $\nu$ , and let  $\mathcal{P}$  be the set of the all problem instances with a unique local best arm at each client and a unique global best arm.

## 2.2 Policy

We assume that the clients are time-synchronised with the central server and communicate with the server at certain pre-defined time instants.

A learner is tasked with the goal of designing an algorithm for finding the local best arm of each client and the global best arm. By an algorithm, we mean a tuple  $\Pi = (\Pi_{\text{comm}}, \Pi_{\text{cli}}, \Pi_{\text{svr}})$  of (a) a strategy for *communication* between the clients and the server, (b) a strategy for selection of arms at each *client*, and (c) a combined stopping and recommendation rule at the *server*. Below, we assume time is discrete and takes values in  $\{1, 2, \dots\}$ .

**Communication Strategy:** The communication strategy  $\Pi_{\text{comm}}$  consists of the following components: (a)  $\{b_r\}_{r \in \mathbb{N}}$ : the time instants of communication, with  $b_r \in \mathbb{N}$  and  $b_r \leq b_{r+1}$  for all  $r \in \mathbb{N}$ , (b)  $\Sigma$ : the set of values transmitted from the server to each client, (c)  $\Phi$ : the set of values transmitted from each client to the server; this is assumed to be *identical* for all the clients, (d)  $h_r : \Phi^{Mr} \rightarrow \Sigma$ : a function deployed at the server, which aggregates the information transmitted from all the clients in the communication rounds  $1, \dots, r$ , and generates the set of output value(s) to be transmitted to each client, and (e)  $\rho_r^m : ([K] \times \mathbb{R})^{b_r} \rightarrow \Phi$ : a function deployed at client  $m \in [M]$ , which aggregates the observations seen locally by client  $m$  from its arms in the time instants  $1, \dots, b_r$ , and generates an output value to be transmitted to the server.

**Arms Selection Strategy:** The arms selection strategy  $\Pi_{\text{cli}}$  consists of component arm selection rules  $\{\pi_t^m : (\mathbb{R} \times [K])^t \times \Sigma \rightarrow \Delta_K : m \in [M], t \geq 1\}$ , where  $\Delta_K$  denotes the probability simplex of dimension  $K - 1$ . Here,  $\pi_t^m$  takes as input the observations seen locally from the arms of client  $m$  up to time  $t$  and the most recent information received from the server to output a probability distribution on  $[K]$  according to which an arm is to be pulled at time  $t + 1$ .

**Stopping and Recommendation Strategy:** The stopping and recommendation strategy  $\Pi_{\text{svr}}$  at the server consists of the following components: (a) the stopping rule  $\nu_r : \Phi^{Mr} \rightarrow \{0, 1\}$  that decides whether the algorithm stops in the  $r$ -th communication round and (b) the recommendation rule  $\Psi_r : \Phi^{Mr} \rightarrow [K]^{M+1}$  to output

the empirical local best arm of each client and the empirical global best arm, if the algorithm stops in the  $r$ th communication round.

We denote a generic policy by  $\pi$ , its stopping time by  $\tau$ , and its recommendations at stoppage by  $\hat{\mathbf{A}} = (\hat{a}_1, \dots, \hat{a}_M, \hat{a})$ .

## 2.3 $\epsilon$ -Differential Privacy

In distributed and federated learning systems, the information held by individual clients is often sensitive and cannot be revealed directly to a central server. Differential privacy, introduced by Dwork and collaborators, provides a rigorous framework for quantifying and controlling the information leakage caused by the output of a randomized algorithm. A mechanism is said to be differentially private if its output distribution remains nearly unchanged when the data of a single individual is modified. That is, an observer cannot reliably infer whether any particular individual's data was used in generating the output.

Let  $\mathcal{D}$  denote the data domain and let  $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{Y}$  be a randomized mechanism. Two datasets  $D, D' \in \mathcal{D}$  are said to be neighbouring if they differ in the data of a single individual.

**Definition 2.1** ( $\epsilon$ -Differential Privacy). *Let  $\epsilon > 0$ . A randomized mechanism  $\mathcal{M}$  is said to satisfy  $\epsilon$ -differential privacy if for every pair of neighbouring datasets  $D$  and  $D'$  and every measurable subset  $S \subseteq \mathcal{Y}$ ,*

$$\Pr(\mathcal{M}(D) \in S) \leq e^\epsilon \Pr(\mathcal{M}(D') \in S).$$

The parameter  $\epsilon$  is referred to as the privacy budget. Smaller values of  $\epsilon$  provide stronger privacy guarantees by requiring the output distributions corresponding to neighbouring datasets to be nearly indistinguishable. Conversely, larger values of  $\epsilon$  permit greater information leakage and typically lead to improved utility.

### 2.3.1 The Laplace Mechanism

One of the most widely used mechanisms for achieving differential privacy is the Laplace mechanism. Consider a function  $f : \mathcal{D} \rightarrow \mathbb{R}^d$  that we wish to release. The sensitivity of  $f$  is defined as



$$\Delta_f = \max_{D \sim D'} \|f(D) - f(D')\|_1,$$

where the maximum is taken over all neighbouring datasets.

The Laplace mechanism releases a noisy version of  $f(D)$  according to

$$\mathcal{M}(D) = f(D) + (Z_1, \dots, Z_d),$$

where the random variables  $Z_i$  are independent and distributed as

$$Z_i \sim \text{Lap}\left(\frac{\Delta_f}{\epsilon}\right).$$

The density of a Laplace random variable with scale parameter  $b$  is given by

$$p(z) = \frac{1}{2b} \exp\left(-\frac{|z|}{b}\right).$$

A fundamental result due to Dwork et al. states that the above mechanism satisfies  $\epsilon$ -differential privacy. The amount of noise added is proportional to the sensitivity of the released quantity and inversely proportional to the privacy budget. Thus, stronger privacy requirements necessitate the addition of larger noise.

### 2.3.2 Privacy Model for Federated Best Arm Identification

We now specialize the notion of differential privacy to the federated best arm identification setting considered in this thesis. In our framework, a central server interacts with  $M$  clients, each of which observes rewards generated by its local environment. Since the rewards may contain sensitive information, the learning procedure must ensure that the information revealed through communication does not compromise the privacy of any individual client.

Consider a generic policy  $\pi$  with stopping time  $\tau$  and recommendation  $\hat{A}$ . Fix  $T \in \mathbb{N}$  and suppose that  $\tau = T$ . Let  $A_T = (A_{t,m} : t \in [T], m \in [M])$  denote the random sequence of arm selections up to time  $T$ , and let  $R_T = (R_{t,m} : t \in [T], m \in [M])$  denote the corresponding sequence of observed rewards. For any deterministic arm sequence  $a_T = (a_{t,m} : t \in [T], m \in [M])$ , deterministic reward sequence  $r_T = (r_{t,m} : t \in [T], m \in [M])$ , and recommendation vector  $\hat{a} \in [K]^{M+1}$ , define

$$\begin{aligned} \pi(T, a_T, \hat{a} \mid r_T) &:= \left[ \prod_{m=1}^M \prod_{t=1}^T P_\nu^\pi \left( A_{t,m} = a_{t,m} \mid A_{1:t-1,m} = a_{1:t-1,m}, R_{1:t-1,m} = r_{1:t-1,m} \right) \right] \\ &\quad \times P_\nu^\pi(\tau = T \mid A_T = a_T, R_T = r_T) \\ &\quad \times P_\nu^\pi(\hat{A} = \hat{a} \mid \tau = T, A_T = a_T, R_T = r_T). \end{aligned}$$

By viewing the policy  $\pi$  as a randomized mechanism whose input is the sequence of rewards and whose output consists of the stopping time, arm selections, and recommendations, the quantity  $\pi(T, a_T, \hat{a} \mid r_T)$  represents the probability of producing the output tuple  $(T, a_T, \hat{a})$  when the reward sequence is  $r_T$ .

**Definition 2.2** ( $\epsilon$ -Global Differential Privacy for Federated BAI). *Let  $\epsilon > 0$ . A policy  $\pi$  is said to satisfy  $\epsilon$ -global differential privacy if, for every  $T \in \mathbb{N}$ , every arm sequence  $a_T$ , every recommendation vector  $\hat{a} \in [K]^{M+1}$ , and every pair of neighbouring reward sequences  $r_T$  and  $\bar{r}_T$ ,*

$$\pi(T, a_T, \hat{a} \mid r_T) \leq e^\epsilon \pi(T, a_T, \hat{a} \mid \bar{r}_T). \quad (2.4)$$

The parameter  $\epsilon$  quantifies the strength of the privacy guarantee. Smaller values of  $\epsilon$  impose stricter privacy requirements by forcing the behavior of the policy to remain nearly unchanged under modifications of a client's reward sequence. Throughout this thesis, all privacy guarantees are understood in the sense of  $\epsilon$ -global differential privacy.

## 2.4 Objective

Given a confidence level  $\delta \in (0, 1)$  and privacy parameter  $\varepsilon > 0$ , let

$$\Pi(\varepsilon, \delta) := \left\{ \pi : \forall \nu \in \mathcal{P}, \quad \mathbb{P}_\nu^\pi(\tau < \infty) = 1, \quad \mathbb{P}_\nu^\pi(\hat{\mathbf{A}} = \mathbf{a}^*(\nu)) \geq 1 - \delta, \quad \pi \text{ satisfies (2.4)} \right\} \quad (2.5)$$

denote the collection of all policies that stop in finite time almost surely, identify the correct set of optimal arms with probability at least  $1 - \delta$ , and satisfy the  $\varepsilon$ -global differential privacy constraint.

The first objective of this thesis is to characterize the fundamental limits of differentially private federated best arm identification. In particular, we seek to determine the instance-dependent sample complexity required by any algorithm belonging to  $\Pi(\varepsilon, \delta)$  and to understand the effect of privacy constraints on the statistical difficulty of the problem. This is captured through the asymptotic quantity

$$\liminf_{\delta \downarrow 0} \inf_{\pi \in \Pi(\varepsilon, \delta)} \frac{\mathbb{E}_\nu^\pi[\tau]}{\log(1/\delta)}, \quad (2.6)$$

which serves as the information-theoretic benchmark for all  $\varepsilon$ -private algorithms.

Building upon this lower bound, the second objective is to design an  $\varepsilon$ -globally differentially private algorithm whose sampling and stopping rules effectively balance statistical efficiency and privacy preservation. The goal is to develop a practical procedure that approaches the performance suggested by the lower-bound analysis while satisfying the required privacy guarantees.

Finally, the proposed algorithm is evaluated through numerical simulations. These experiments are used to study the impact of the privacy parameter on sample complexity, verify the correctness of the algorithm, and compare the observed empirical behavior with the theoretical predictions obtained from the lower-bound analysis.

In (2.5) and (2.6),  $\mathbb{P}_\nu^\pi$  and  $\mathbb{E}_\nu^\pi$  denote probabilities and expectations computed under the bandit instance  $\nu$  and policy  $\pi$ , respectively.

# Chapter 3

## Lower Bound Analysis

### 3.1 Lower Bound

In this section, we present a lower bound on the limiting growth rate of the expected time required to find the local best arms and the global best arm as  $\delta \downarrow 0$ . For any  $x, y \in [0, 1]$ , we write  $\text{kl}(x, y)$  to denote the binary relative entropy function, defined as

$$\text{kl}(x, y) \coloneqq x \log \frac{x}{y} + (1 - x) \log \frac{1 - x}{1 - y}.$$

Furthermore, for any two probability distributions  $P$  and  $Q$  defined on the same alphabet, we write  $\text{KL}(P||Q)$  to denote the relative entropy (or Kullback–Leibler divergence) between the distributions  $P$  and  $Q$ .

**Proposition 3.1.** *Fix  $\varepsilon > 0$ . Under the instance  $\nu \in \mathcal{P}$ ,*

$$\liminf_{\delta \downarrow 0} \inf_{\pi \in \Pi(\varepsilon, \delta)} \frac{\mathbb{E}_{\nu}^{\pi}[\tau]}{\log(1/\delta)} \geq \frac{1}{\mathcal{K}^*(\nu, \varepsilon)}, \quad (3.1)$$

where the constant  $\mathcal{K}^*(\nu, \varepsilon)$  is given by

$$\mathcal{K}^*(\nu, \varepsilon) \coloneqq \sup_{\omega \in \Delta_K^M} \inf_{\nu' \in \text{ALT}(\nu)} \min \left\{ 6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{TV}(\nu_{a,m} || \nu'_{a,m}), \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{KL}(\nu_{a,m} || \nu'_{a,m}) \right\}. \quad (3.2)$$

In (3.2), the outer supremum is over  $M$ -fold copies of probability distributions on the arms, one for each client.

The derivation follows the classical change-of-measure argument of [1], combined with the transportation inequality for differentially private algorithms established by

[3]. Let  $\nu$  denote the true bandit instance and let  $\nu' \in \text{ALT}(\nu)$  be an alternative instance whose optimal arm differs from that of  $\nu$ .

Define the event  $E = \{\hat{A} = a^*(\nu)\}$ , where  $a^*(\nu) = \arg \max_{a \in [K]} \mu_a$  denotes the optimal arm under the true instance. Since the policy is  $\delta$ -PAC, we have

$$\mathbb{P}_\nu(E) \geq 1 - \delta, \quad \mathbb{P}_{\nu'}(E) \leq \delta.$$

We begin by introducing the log-likelihood ratio between the instances  $\nu$  and  $\nu'$ :

$$L_t(\nu, \nu') = \sum_{m=1}^M \sum_{a=1}^K \sum_{s=1}^{N_{a,m}(t)} \log \frac{f_{\nu_{a,m}}(X_{a,m,s})}{f_{\nu'_{a,m}}(X_{a,m,s})}.$$

Using Wald's lemma together with the independence of rewards across clients and arms yields

$$\mathbb{E}_\nu [L_\tau(\nu, \nu')] = \sum_{m=1}^M \sum_{a=1}^K \mathbb{E}_\nu [N_{a,m}(\tau)] \text{KL}(\nu_{a,m} \| \nu'_{a,m}).$$

Applying the data-processing inequality to the indicator random variable  $\mathbf{1}_E$ , exactly as in the proof of Garivier and Kaufmann [1], gives

$$\mathbb{E}_\nu [L_\tau(\nu, \nu')] \geq \text{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E)).$$

Combining the previous two displays and using the  $\delta$ -PAC property yields

$$\sum_{m=1}^M \sum_{a=1}^K \mathbb{E}_\nu [N_{a,m}(\tau)] \text{KL}(\nu_{a,m} \| \nu'_{a,m}) \geq \text{kl}(1 - \delta, \delta).$$

This is the classical information-theoretic inequality underlying the lower bound for best arm identification.

The privacy constraint introduces an additional limitation on the amount of information that can be extracted from the observations. In particular, Azize et al. [3] establish a transportation inequality for  $\varepsilon$ -globally differentially private bandit algorithms by combining the Sequential Karwa-Vadhan lemma with a data-processing argument adapted to the sequential setting. Extending their argument to the federated setting by summing over all client-arm pairs yields

$$6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \mathbb{E}_\nu [N_{a,m}(\tau)] \text{TV}(\nu_{a,m} \parallel \nu'_{a,m}) \geq \text{kl}(1 - \delta, \delta).$$

The above inequality captures the contraction in distinguishability induced by the privacy mechanism. Unlike the classical lower bound, which depends on Kullback–Leibler divergence, the privacy constraint forces the information gain to be controlled through total variation distance.

Since both inequalities hold simultaneously, we obtain

$$\min \left\{ 6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \mathbb{E}_\nu [N_{a,m}(\tau)] \text{TV}(\nu_{a,m} \parallel \nu'_{a,m}), \right. \\ \left. \sum_{m=1}^M \sum_{a=1}^K \mathbb{E}_\nu [N_{a,m}(\tau)] \text{KL}(\nu_{a,m} \parallel \nu'_{a,m}) \right\} \geq \text{kl}(1 - \delta, \delta).$$

Define the normalized sampling proportions

$$\omega_{a,m} = \frac{\mathbb{E}_\nu [N_{a,m}(\tau)]}{\mathbb{E}_\nu [\tau]}.$$

Dividing the previous inequality by  $\mathbb{E}_\nu [\tau]$  gives

$$\mathbb{E}_\nu [\tau] \min \left\{ 6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{TV}(\nu_{a,m} \parallel \nu'_{a,m}), \right. \\ \left. \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{KL}(\nu_{a,m} \parallel \nu'_{a,m}) \right\} \geq \frac{\text{kl}(1 - \delta, \delta)}{\mathbb{E}_\nu [\tau]}.$$

Rearranging yields

$$\mathbb{E}_\nu [\tau] \geq \frac{\text{kl}(1 - \delta, \delta)}{\min \left\{ 6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{TV}(\nu_{a,m} \parallel \nu'_{a,m}), \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{KL}(\nu_{a,m} \parallel \nu'_{a,m}) \right\}}.$$

Since the above inequality holds for every alternative instance  $\nu' \in \text{ALT}(\nu)$ , we may take the infimum over all alternatives:

$$\mathbb{E}_\nu[\tau] \geq \frac{\text{kl}(1 - \delta, \delta)}{\inf_{\nu' \in \text{ALT}(\nu)} \min \left\{ 6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{TV}(\nu_{a,m} \| \nu'_{a,m}), \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{KL}(\nu_{a,m} \| \nu'_{a,m}) \right\}}.$$

Finally, maximizing over all feasible allocation vectors  $\omega \in \Delta_K^M$  gives

$$\mathbb{E}_\nu[\tau] \geq \frac{\text{kl}(1 - \delta, \delta)}{\sup_{\omega \in \Delta_K^M} \inf_{\nu' \in \text{ALT}(\nu)} \min \left\{ 6\varepsilon \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{TV}(\nu_{a,m} \| \nu'_{a,m}), \sum_{m=1}^M \sum_{a=1}^K \omega_{a,m} \text{KL}(\nu_{a,m} \| \nu'_{a,m}) \right\}}.$$

Recognizing the denominator as  $\mathcal{K}^*(\nu, \varepsilon)$ , we obtain

$$\mathbb{E}_\nu[\tau] \geq \frac{\text{kl}(1 - \delta, \delta)}{\mathcal{K}^*(\nu, \varepsilon)}.$$

Using the asymptotic relation

$$\text{kl}(1 - \delta, \delta) = (1 + o(1)) \log \frac{1}{\delta}, \quad \delta \downarrow 0,$$

we conclude that

$$\liminf_{\delta \downarrow 0} \inf_{\pi \in \Pi(\varepsilon, \delta)} \frac{\mathbb{E}_\nu^\pi[\tau]}{\log(1/\delta)} \geq \frac{1}{\mathcal{K}^*(\nu, \varepsilon)},$$

which establishes Proposition 3.1.

## 3.2 Reduction and analysis of the lower bound

Restrict to Bernoulli rewards. For  $p, q \in (0, 1)$ ,

$$\text{TV}(\text{Ber}(p), \text{Ber}(q)) = |p - q|, \quad \text{KL}(\text{Ber}(p), \text{Ber}(q)) = d(p, q) := p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q},$$

so the bound depends on the instance only through the means  $\mu = (\mu_{a,m})$ . Write  $\bar{\mu}_a = \frac{1}{M} \sum_m \mu_{a,m}$ , the local optima  $b_m = \argmax_a \mu_{a,m}$ , the global optimum  $a^* = \argmax_a \bar{\mu}_a$ , and the global gap  $\Delta_a = \bar{\mu}_{a^*} - \bar{\mu}_a$ . The allocation  $\omega = (\omega_{a,m})$  ranges

over  $\Delta_K^M$  (one simplex per client). The characteristic quantity is

$$\mathcal{K}^*(\nu, \varepsilon) = \sup_{\omega \in \Delta_K^M} \inf_{\nu' \in \text{Alt}(\nu)} \min \left\{ 6\varepsilon \sum_{m,a} \omega_{a,m} |\mu_{a,m} - \mu'_{a,m}|, \sum_{m,a} \omega_{a,m} d(\mu_{a,m}, \mu'_{a,m}) \right\}.$$

### 3.2.1 Separating privacy from information

**Lemma 3.2.** *For any two functions  $f, g$  on a common domain,*

$$\inf_x \min\{f(x), g(x)\} = \min \left\{ \inf_x f(x), \inf_x g(x) \right\}.$$

*Proof.* Since  $\min\{f(x), g(x)\} \leq f(x)$ , we have

$$\inf_x \min\{f(x), g(x)\} \leq \inf_x f(x),$$

and likewise

$$\inf_x \min\{f(x), g(x)\} \leq \inf_x g(x).$$

Therefore,

$$\inf_x \min\{f(x), g(x)\} \leq \min \left\{ \inf_x f(x), \inf_x g(x) \right\}.$$

Conversely,  $f(x) \geq \inf_x f(x)$  and  $g(x) \geq \inf_x g(x)$  imply that

$$\min\{f(x), g(x)\} \geq \min \left\{ \inf_x f(x), \inf_x g(x) \right\}$$

for every  $x$ . Taking the infimum over  $x$  on the left-hand side yields the reverse inequality.  $\square$

The inner objective is

$$\min\{f, g\},$$

with

$$f = 6\varepsilon \sum_{m,a} \omega_{a,m} |\mu_{a,m} - \mu'_{a,m}|$$

and

$$g = \sum_{m,a} \omega_{a,m} d(\mu_{a,m}, \mu'_{a,m}).$$

The constant  $6\varepsilon > 0$  passes through the infimum, so Lemma 3.2 gives



$$\begin{aligned}
 \mathcal{K}^*(\nu, \varepsilon) &= \sup_{\omega \in \Delta_K^M} \min\{6\varepsilon T(\omega), D(\omega)\}, \\
 T(\omega) &= \inf_{\nu'} \sum_{m,a} \omega_{a,m} |\mu_{a,m} - \mu'_{a,m}|, \\
 D(\omega) &= \inf_{\nu'} \sum_{m,a} \omega_{a,m} d(\mu_{a,m}, \mu'_{a,m}).
 \end{aligned}$$

The privacy term  $T$  and the information term  $D$  are now independent minimizations.

### 3.2.2 Decomposing the alternative set

The set  $\text{ALT}(\nu)$  consists of all instances whose optimal tuple differs from that of  $\nu$ . This can occur either when a local optimal arm  $b_m$  changes at some client  $m$ , or when the global optimal arm  $a^*$  changes. Consequently, the alternative set can be decomposed as

$$\text{ALT}(\nu) = \left( \bigcup_{m \in [M]} \bigcup_{a \neq b_m} \mathcal{E}_{m,a} \right) \cup \left( \bigcup_{a \neq a^*} \mathcal{G}_a \right).$$

Here,

$$\mathcal{E}_{m,a} = \left\{ \nu' : \mu'_{a,m} \geq \mu'_{b_m,m} \right\},$$

which corresponds to promoting arm  $a$  over the current local best arm  $b_m$  at client  $m$ , and

$$\mathcal{G}_a = \left\{ \nu' : \sum_{m=1}^M \mu'_{a,m} \geq \sum_{m=1}^M \mu'_{a^*,m} \right\},$$

which corresponds to promoting arm  $a$  to become the globally optimal arm.

Both  $T$  and  $D$  minimize a sum of nonnegative per-coordinate costs, so the infimum over a union is the minimum of the per-event infima:

$$T(\omega) = \min \left\{ \min_{\substack{m \in [M] \\ a \neq b_m}} T_{m,a}^{\text{loc}}(\omega), \min_{a \neq a^*} T_a^{\text{glob}}(\omega) \right\},$$

$$D(\omega) = \min \left\{ \min_{\substack{m \in [M] \\ a \neq b_m}} D_{m,a}^{\text{loc}}(\omega), \min_{a \neq a^*} D_a^{\text{glob}}(\omega) \right\}.$$

Each event is solved separately below; the binding event sets the value.

### 3.2.3 Local events

Consider a local confusion event  $\mathcal{E}_{m,a}$ , in which arm  $a$  overtakes the current local best arm  $b_m$  at client  $m$ . By construction, the event only involves the two coordinates  $(b_m, m)$  and  $(a, m)$ , while all other means remain unchanged. Since both the KL and TV objectives are sums of nonnegative coordinate-wise costs, the cheapest alternative instance is obtained by perturbing only these two coordinates until they meet at a common value  $c$ .

**KL term.** The corresponding optimization problem is

$$D_{m,a}^{\text{loc}}(\omega) = \inf_{c \in [0,1]} \left[ \omega_{b_m,m} d(\mu_{b_m,m}, c) + \omega_{a,m} d(\mu_{a,m}, c) \right].$$

To determine the optimal meeting point, we differentiate the objective with respect to  $c$ . For Bernoulli distributions,

$$\frac{\partial}{\partial c} d(p, c) = \frac{c - p}{c(1 - c)}.$$

Therefore, the first-order optimality condition becomes

$$\omega_{b_m,m} \frac{c - \mu_{b_m,m}}{c(1 - c)} + \omega_{a,m} \frac{c - \mu_{a,m}}{c(1 - c)} = 0.$$

Since  $c \in (0, 1)$ , the denominator  $c(1 - c)$  is strictly positive and can be cancelled, yielding

$$\omega_{b_m,m}(c - \mu_{b_m,m}) + \omega_{a,m}(c - \mu_{a,m}) = 0.$$

Solving for  $c$  gives

$$c^* = \frac{\omega_{b_m,m} \mu_{b_m,m} + \omega_{a,m} \mu_{a,m}}{\omega_{b_m,m} + \omega_{a,m}}.$$

Thus, the optimal alternative is obtained by moving both means to their weighted arithmetic average. Substituting  $c^*$  back into the objective yields

$$D_{m,a}^{\text{loc}}(\omega) = \omega_{b_m,m} d(\mu_{b_m,m}, c^*) + \omega_{a,m} d(\mu_{a,m}, c^*).$$

**TV term.** For the total variation component, the local optimization problem takes the form

$$T_{m,a}^{\text{loc}}(\omega) = \inf_{c \in [0,1]} \left[ \omega_{b_m,m} |\mu_{b_m,m} - c| + \omega_{a,m} |\mu_{a,m} - c| \right].$$

This is a weighted median problem on the interval joining the two means. The objective is piecewise linear, and its minimum is attained at the endpoint carrying the larger weight. Evaluating the objective at the optimum yields the explicit expression

$$T_{m,a}^{\text{loc}}(\omega) = \min(\omega_{b_m,m}, \omega_{a,m}) |\mu_{b_m,m} - \mu_{a,m}|.$$

The TV term is proportional to the local mean gap and weighted by the smaller of the two sampling proportions.

### 3.2.4 Global events

We now consider a global confusion event  $\mathcal{G}_a$ , in which a suboptimal arm  $a$  becomes globally optimal. Unlike the local events studied previously, the global constraint couples all clients simultaneously. In particular, arm  $a$  overtakes the true global best arm  $a^*$  when

$$\sum_{m=1}^M \mu'_{a,m} = \sum_{m=1}^M \mu'_{a^*,m}.$$

Consequently, the optimization is no longer restricted to two coordinates. Instead, all  $2M$  coordinates

$$\{(a^*, m), (a, m)\}_{m=1}^M$$

participate in the perturbation, making the problem more involved than its local counterpart.

**KL term.** For a fixed allocation vector  $\omega$ , the KL contribution associated with the event  $\mathcal{G}_a$  is obtained by solving

$$D_a^{\text{glob}}(\omega) = \inf_{\mu'} \sum_{m=1}^M \left[ \omega_{a^*,m} d(\mu_{a^*,m}, \mu'_{a^*,m}) + \omega_{a,m} d(\mu_{a,m}, \mu'_{a,m}) \right]$$

subject to

$$\sum_{m=1}^M \mu'_{a,m} = \sum_{m=1}^M \mu'_{a^*,m}.$$

Introducing a Lagrange multiplier  $\lambda$  for the equality constraint yields the Lagrangian

$$L(\mu', \lambda) = \sum_{m=1}^M \left[ \omega_{a^*,m} d(\mu_{a^*,m}, \mu'_{a^*,m}) + \omega_{a,m} d(\mu_{a,m}, \mu'_{a,m}) \right] + \lambda \left( \sum_{m=1}^M \mu'_{a,m} - \sum_{m=1}^M \mu'_{a^*,m} \right).$$

Using

$$\frac{\partial}{\partial q} d(p, q) = \frac{q - p}{q(1 - q)},$$

the first-order optimality conditions become

$$\omega_{a,m} \frac{\mu'_{a,m} - \mu_{a,m}}{\mu'_{a,m}(1 - \mu'_{a,m})} = \lambda, \quad m \in [M],$$

and

$$\omega_{a^*,m} \frac{\mu'_{a^*,m} - \mu_{a^*,m}}{\mu'_{a^*,m}(1 - \mu'_{a^*,m})} = -\lambda, \quad m \in [M].$$

The crucial difference from the local case is that the denominators no longer cancel. In the local event, both coordinates shared the same meeting point, causing a common factor to disappear. Here, all  $2M$  coordinates are coupled through a single multiplier  $\lambda$ , and each coordinate must satisfy its own nonlinear equation.

Nevertheless, for a fixed value of  $\lambda$ , each equation admits a unique solution because the left-hand side is monotone in the corresponding variable. Hence every coordinate can be expressed as a function of  $\lambda$ ,

$$\mu'_{a,m} = \mu'_{a,m}(\lambda), \quad \mu'_{a^*,m} = \mu'_{a^*,m}(\lambda).$$

Substituting these expressions into the equality constraint reduces the problem to a one-dimensional root-finding problem:

$$\sum_{m=1}^M \left( \mu'_{a,m}(\lambda) - \mu'_{a^*,m}(\lambda) \right) = 0.$$

The multiplier can therefore be computed efficiently using bisection or Newton's method. Once the optimal multiplier is obtained, the value of the objective yields  $D_a^{\text{glob}}(\omega)$ . The problem is convex and can be solved numerically with negligible computational cost.

**TV term.** The total variation component exhibits a different structure. Since

$$\text{TV}(\text{Ber}(p), \text{Ber}(q)) = |p - q|,$$

the corresponding optimization problem becomes

$$T_a^{\text{glob}}(\omega) = \inf_{\mu'} \sum_{m=1}^M \left[ \omega_{a^*,m} |\mu'_{a^*,m} - \mu_{a^*,m}| + \omega_{a,m} |\mu'_{a,m} - \mu_{a,m}| \right]$$

subject to

$$\sum_{m=1}^M \mu'_{a,m} = \sum_{m=1}^M \mu'_{a^*,m},$$

and

$$0 \leq \mu'_{a,m}, \mu'_{a^*,m} \leq 1.$$

Define the global gap

$$G_a = \sum_{m=1}^M (\mu_{a^*,m} - \mu_{a,m}) = M\Delta_a.$$

The constraint requires a total displacement of magnitude  $G_a$  to be transferred between the two arm blocks. Since the objective is linear in the perturbations, the optimal strategy is to allocate this displacement to the coordinates with the smallest weights first. This is precisely the structure of a fractional knapsack problem with box constraints.

As long as none of the coordinates reaches the boundaries 0 or 1, the displacement is absorbed entirely by the least expensive coordinate, leading to the approximation  $T_a^{\text{glob}}(\omega) \approx M\Delta_a \cdot \min_{(c,m)} \omega_{c,m}$ .

When box constraints become active, the displacement is redistributed to the next cheapest coordinate, exactly as in the standard fractional knapsack procedure. Although a simple closed-form expression is generally unavailable, the optimization remains straightforward and can be solved efficiently by sorting the coordinates according to their weights.

The global KL and TV optimizations therefore differ substantially in their analytical form, yet both retain a tractable convex structure. This observation is important for the numerical computation of the characteristic time, since each global confusion event can be evaluated efficiently despite the absence of an explicit formula.

### 3.2.5 Concavity of the optimization problem

For a fixed alternative instance  $\nu'$ , both

$$\sum_{m,a} \omega_{a,m} \text{TV}(\mu_{a,m}, \mu'_{a,m})$$

and

$$\sum_{m,a} \omega_{a,m} d(\mu_{a,m}, \mu'_{a,m})$$

are affine functions of the allocation vector  $\omega$ . The infimum of a family of affine functions in  $\omega$  is concave. Taking a minimum over the finite collection of local and global confusion events preserves concavity as well.

Define

$$F(\omega) = \min \left\{ \min_{\substack{m \in [M] \\ a \neq b_m}} \min \left\{ 6\varepsilon T_{m,a}^{\text{loc}}(\omega), D_{m,a}^{\text{loc}}(\omega) \right\}, \min_{a \neq a^*} \min \left\{ 6\varepsilon T_a^{\text{glob}}(\omega), D_a^{\text{glob}}(\omega) \right\} \right\}.$$

It follows that  $F(\omega)$  is concave on the feasible set  $\Delta_K^M$ . Therefore,

$$\mathcal{K}^*(\nu, \varepsilon) = \sup_{\omega \in \Delta_K^M} F(\omega)$$

is a concave maximization problem over a product of simplices. Every local optimum is globally optimal, and the set of maximizers is convex.

The local inner optimization problems admit closed-form solutions, while the global inner problems reduce respectively to a one-dimensional root-finding procedure for the KL term and a fractional-knapsack computation for the TV term. As a result, each outer optimization step is computationally inexpensive.

### 3.3 Two Privacy Regimes

The simplified lower bound obtained in the previous section is

$$\mathcal{K}^*(\nu, \varepsilon) = \sup_{\omega \in \Delta_K^M} \min \{6\varepsilon T(\omega), D(\omega)\}.$$

Therefore, every  $\varepsilon$ -DP algorithm satisfies

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\nu[\tau_\delta]}{\log(1/\delta)} \geq \frac{1}{\mathcal{K}^*(\nu, \varepsilon)}.$$

An important feature of this expression is that the privacy parameter  $\varepsilon$  appears only as a multiplicative factor in front of the total variation term. The quantities  $T(\omega)$  and  $D(\omega)$  depend only on the underlying instance and the allocation vector, whereas the privacy budget merely determines which of the two branches inside the minimum is active.

For a fixed allocation  $\omega$ , define

$$F_\omega(\varepsilon) = \min \{6\varepsilon T(\omega), D(\omega)\}.$$

The function  $F_\omega(\varepsilon)$  is monotone increasing and piecewise linear. Indeed,

$$F_\omega(\varepsilon) = 6\varepsilon T(\omega), \quad \text{whenever } \varepsilon \leq \frac{D(\omega)}{6T(\omega)},$$

and

$$F_\omega(\varepsilon) = D(\omega), \quad \text{whenever } \varepsilon \geq \frac{D(\omega)}{6T(\omega)}.$$

Thus, for small privacy budgets the total variation branch is active, whereas for sufficiently large privacy budgets the KL branch becomes active. Since taking the supremum over  $\omega$  preserves monotonicity, the characteristic quantity  $\mathcal{K}^*(\nu, \varepsilon)$  is itself non-decreasing in  $\varepsilon$  and satisfies

$$0 \leq \mathcal{K}^*(\nu, \varepsilon) \leq \mathcal{K}_{\text{NP}}^*,$$

where

$$\mathcal{K}_{\text{NP}}^* = \sup_{\omega \in \Delta_K^M} D(\omega)$$

is the characteristic quantity associated with the corresponding non-private problem.

The behaviour of the lower bound is governed by a critical privacy level at which the TV and KL branches intersect.

**Proposition 3.3** (Privacy regimes). *Let*

$$\omega_T \in \arg \max_{\omega \in \Delta_K^M} T(\omega), \quad \omega_D \in \arg \max_{\omega \in \Delta_K^M} D(\omega),$$

and define

$$\varepsilon^* = \frac{D(\omega_D)}{6T(\omega_T)} = \frac{\mathcal{K}_{\text{NP}}^*}{6 \sup_{\omega} T(\omega)}.$$

Then the following two regimes arise.

**1. Privacy-dominated regime** ( $\varepsilon \lesssim \varepsilon^*$ ).

*The total variation branch is active at the optimum and*

$$\mathcal{K}^*(\nu, \varepsilon) = 6\varepsilon \sup_{\omega \in \Delta_K^M} T(\omega).$$



Consequently,

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\nu[\tau_\delta]}{\log(1/\delta)} = \Theta\left(\frac{1}{\varepsilon}\right).$$

**2. Information-dominated regime** ( $\varepsilon \gtrsim \varepsilon^*$ ).

The KL branch becomes active and

$$\mathcal{K}^*(\nu, \varepsilon) = \sup_{\omega \in \Delta_K^M} D(\omega) = \mathcal{K}_{\text{NP}}^*.$$

Hence the privacy constraint no longer affects the asymptotic sample complexity and the classical non-private lower bound is recovered.

*Proof.* For every allocation vector  $\omega$ ,

$$\min \{6\varepsilon T(\omega), D(\omega)\} \leq 6\varepsilon T(\omega),$$

which implies

$$\mathcal{K}^*(\nu, \varepsilon) \leq 6\varepsilon \sup_{\omega} T(\omega).$$

Evaluating the objective at  $\omega_T$  shows that equality holds whenever

$$6\varepsilon T(\omega_T) \leq D(\omega_T).$$

This yields the privacy-dominated regime.

Similarly,

$$\min \{6\varepsilon T(\omega), D(\omega)\} \leq D(\omega),$$

and therefore

$$\mathcal{K}^*(\nu, \varepsilon) \leq \sup_{\omega} D(\omega).$$

Evaluating the objective at  $\omega_D$  shows that equality holds whenever

$$6\varepsilon T(\omega_D) \geq D(\omega_D).$$

This yields the information-dominated regime. The transition occurs when the two bounds coincide, giving the threshold  $\varepsilon^*$ .  $\square$

Proposition 3.3 reveals a sharp privacy–information trade-off. The KL term measures the statistical information available for distinguishing the true instance from competing alternatives, while the TV term quantifies the information loss imposed by differential privacy. Since

$$\min \{6\varepsilon T(\omega), D(\omega)\} \leq D(\omega),$$

privacy can only reduce the effective information available to the learner.

When  $\varepsilon < \varepsilon^*$ , the privacy constraint is the dominant bottleneck. In this regime the characteristic quantity scales linearly with  $\varepsilon$ , implying that the sample complexity scales as  $1/\varepsilon$ . Thus, halving the privacy budget approximately doubles the number of samples required for reliable identification.

When  $\varepsilon > \varepsilon^*$ , the privacy budget is sufficiently large that the privacy constraint ceases to be active. The achievable information is then limited solely by the statistical difficulty of the underlying instance, and the asymptotic complexity coincides with that of the non-private problem.

Therefore, if one plots the asymptotic sample complexity as a function of  $\varepsilon$  on a logarithmic scale, one expects to observe a characteristic “knee” at  $\varepsilon^*$ : a  $1/\varepsilon$  decay in the privacy-dominated regime followed by a plateau corresponding to the non-private complexity. This transition point serves as a useful empirical signature of the privacy–information trade-off and will be illustrated in the numerical experiments presented later.

# Chapter 4

## Algorithm Design

In the previous chapter, we established an instance-dependent lower bound for the federated differentially private best arm identification problem and studied its structure under Bernoulli rewards. The lower bound revealed two competing sources of difficulty. The first is the statistical difficulty of distinguishing the true instance from alternative instances, captured by the KL divergence term. The second is the information loss introduced by the differential privacy constraint, captured by the total variation term. Motivated by this characterization, we now develop an algorithm whose design explicitly accounts for both aspects.

The proposed algorithm, referred to as FED-DP-TT (Federated Differentially Private Top-Two Sampling), combines three key ingredients. First, privacy is enforced through episodic releases of privatized empirical means using the Laplace mechanism. Second, sampling decisions are guided by uncertainty in the spirit of Top-Two sampling. Finally, generalized likelihood ratio statistics are employed to determine when sufficient evidence has been accumulated for both local and global decisions.

The objective of the algorithm is to identify

1. the best arm at every client, and
2. the globally best arm obtained by averaging rewards across clients,

while satisfying  $\varepsilon$ -global differential privacy and minimizing the stopping time.

### 4.1 Privacy Preserving Architecture

The privacy mechanism is implemented entirely at the client side. Clients never transmit raw rewards to the server. Instead, only privatized statistics are communicated.

The remaining operations performed by the server consist solely of post-processing and therefore incur no additional privacy cost.

The privacy layer of the algorithm consists of three components:

1. arm-dependent doubling episodes,
2. a forgetting mechanism,
3. Laplace noise injection.

Together, these ingredients guarantee  $\varepsilon$ -global differential privacy while ensuring that the variance introduced by the privacy mechanism vanishes asymptotically.

### 4.1.1 Doubling Episodes

For every client  $m$  and arm  $a$ , observations are grouped into episodes. Unlike fixed-length batching schemes, episode lengths increase geometrically.

Let  $N_{a,m}^{(r)}$  denote the cumulative number of pulls of arm  $a$  at client  $m$  at the end of episode  $r$ . The episode closes whenever the cumulative count doubles, that is,

$$N_{a,m}^{(r+1)} = 2N_{a,m}^{(r)}.$$

Therefore, the episode lengths evolve as

$$1, 2, 4, 8, \dots$$

Whenever an episode closes, the client computes a privatized estimate of the mean reward collected during that episode and transmits it to the server. Since the episode lengths grow geometrically, the number of releases generated by any arm after  $T$  pulls is bounded by

$$O(\log T).$$

Thus, communication frequency grows only logarithmically with the time horizon. This reduction in communication plays an important role in controlling both privacy leakage and communication overhead.

### 4.1.2 Forgetting Mechanism

A fundamental difficulty in repeatedly releasing empirical means is that consecutive releases may depend on overlapping data. In such situations, privacy losses accumulate through sequential composition.

To avoid this issue, we adopt a forgetting mechanism. Let

$$\mathcal{D}_{a,m}^{(r)}$$

denote the set of rewards collected during episode  $r$  of arm  $a$  at client  $m$ . The empirical mean released at the end of the episode is computed exclusively from the rewards contained in  $\mathcal{D}_{a,m}^{(r)}$ .

After the statistic is released, the episode buffer is discarded and a new episode begins. Thus,

$$\mathcal{D}_{a,m}^{(r)} \cap \mathcal{D}_{a,m}^{(s)} = \emptyset, \quad r \neq s.$$

So, every release is computed from a disjoint set of observations. The privacy cost therefore follows from parallel composition rather than sequential composition and the privacy guarantee does not deteriorate with the number of communication rounds.

This observation is crucial because the algorithm may run for a very large number of samples while still maintaining a fixed privacy budget.

### 4.1.3 Laplace Noise Injection

At the end of an episode containing  $n$  observations, client  $m$  computes the empirical mean

$$\hat{\mu}_{a,m} = \frac{1}{n} \sum_{t \in \mathcal{D}_{a,m}^{(r)}} R_{t,a,m}.$$

Since rewards are Bernoulli, they belong to the interval  $[0, 1]$ . Therefore the  $\ell_1$  sensitivity of the empirical mean is

$$\Delta_1 = \frac{1}{n}.$$

Applying the Laplace mechanism, the released statistic is

$$\tilde{\mu}_{a,m} = \hat{\mu}_{a,m} + Z, \quad Z \sim \text{Lap}\left(\frac{1}{n\varepsilon}\right).$$

By the standard privacy guarantee of the Laplace mechanism, each release is  $\varepsilon$ -differentially private.

The server never observes the empirical mean itself and receives only the privatized quantity  $\tilde{\mu}_{a,m}$ .

An important property of the mechanism is that the variance of the added noise decreases as the episode length increases. Specifically,

$$\text{Var}(Z) = \frac{2}{n^2\varepsilon^2}.$$

Hence, as the episode lengths grow geometrically, the effect of the privacy noise becomes progressively smaller. This behavior mirrors the high-privacy regime identified in the lower bound analysis, where the sample complexity scales inversely with the privacy parameter  $\varepsilon$ .

## 4.2 Communication Protocol

Communication between the clients and the server is event-driven rather than periodic. A communication round is triggered whenever at least one arm completes an episode and produces a new privatized estimate. This design significantly reduces the communication burden compared to synchronous federated learning algorithms, where communication occurs after every local update.

At the end of an episode, client  $m$  transmits the tuple

$$\left(\tilde{\mu}_{a,m}, \tilde{N}_{a,m}, \gamma_{\text{loc}}^{(m)}\right),$$

where  $\tilde{\mu}_{a,m}$  denotes the privatized empirical mean,  $\tilde{N}_{a,m}$  denotes the number of pulls of arm  $a$  in the last phase, and  $\gamma_{\text{loc}}^{(m)}$  is the local confidence ratio associated with client  $m$ .

Using the information received from all clients, the server updates the global statistics and broadcasts

$$(B, C, \gamma_{\text{glob}})$$

to all clients, where  $B$  denotes the current global leader,  $C$  denotes the current global challenger, and  $\gamma_{\text{glob}}$  denotes the confidence ratio associated with the global best-arm identification problem.

Since communication only occurs when episode lengths double, the total number of communication rounds generated by any arm after  $T$  pulls is at most  $O(\log T)$ . As a result, the overall communication complexity grows logarithmically with the sample complexity of the algorithm.

All quantities transmitted to the server are already privatized. Hence the communication protocol itself does not introduce any additional privacy leakage beyond that incurred by the Laplace mechanism at the clients.

### 4.3 Local Best Arm Identification

We first describe how each client identifies its own best arm. The local identification problem closely follows the philosophy of Top-Two sampling. At every stage, each client maintains a leader arm that currently appears optimal and a challenger arm that represents the most plausible alternative.

The objective of the local sampling rule is to gather evidence that separates the leader from its strongest competitor while simultaneously accounting for the uncertainty introduced by the privacy mechanism.

#### 4.3.1 Local Leader Selection

Let  $\tilde{\mu}_{a,m}$  denote the most recent privatized estimate of the mean reward of arm  $a$  at client  $m$ , and let  $\tilde{N}_{a,m}$  denote the corresponding pull count.

For each arm we define the privacy-aware upper confidence index

$$U_{a,m} = \tilde{\mu}_{a,m} + \sqrt{\frac{k_{a,m}}{\tilde{N}_{a,m}}} + \frac{k_{a,m}}{\varepsilon \tilde{N}_{a,m}},$$

where

$$k_{a,m} = \log_2(\tilde{N}_{a,m}).$$

The first exploration term corresponds to the usual statistical uncertainty associated with empirical averages. The second term compensates for the additional uncertainty introduced by the Laplace noise.

The local leader is chosen as

$$b_m = \arg \max_{a \in [K]} U_{a,m}.$$

Intuitively, the leader is the arm that currently has the strongest optimistic estimate after accounting for both statistical and privacy-related uncertainty.

### 4.3.2 Local Challenger Selection

Once the leader  $b_m$  has been identified, the next task is to determine the arm that constitutes the most credible alternative.

Following the Top-Two principle, the challenger is defined as the arm that minimizes the generalized likelihood ratio against the leader. Formally,

$$c_m = \arg \min_{a \neq b_m} \text{GLR}_m(b_m, a).$$

Unlike classical UCB algorithms, which focus exclusively on the empirically best arm, the challenger explicitly identifies the arm that is hardest to distinguish from the leader.

Thus, the pair  $(b_m, c_m)$  represents the most informative local comparison at client  $m$ .

### 4.3.3 Local Generalized Likelihood Ratio

The confidence in the local leader is quantified through a generalized likelihood ratio statistic.

For two candidate arms  $b$  and  $a$ , define

$$c = \frac{\tilde{N}_{b,m} \tilde{\mu}_{b,m} + \tilde{N}_{a,m} \tilde{\mu}_{a,m}}{\tilde{N}_{b,m} + \tilde{N}_{a,m}}.$$

The quantity  $c$  corresponds to the empirical mean under the alternative hypothesis that both arms have identical means.

The local GLR statistic is then

$$\text{GLR}_m(b, a) = \tilde{N}_{b,m} d(\tilde{\mu}_{b,m}, c) + \tilde{N}_{a,m} d(\tilde{\mu}_{a,m}, c),$$

where



$$d(x, y) = x \log\left(\frac{x}{y}\right) + (1 - x) \log\left(\frac{1 - x}{1 - y}\right)$$

denotes the Bernoulli Kullback–Leibler divergence.

The statistic measures the amount of evidence against the hypothesis that the two arms have equal means. Larger values indicate stronger evidence in favour of the leader.

### 4.3.4 Local Confidence Ratio

The local stopping rule is expressed through a normalized confidence ratio.

Let  $\beta_{\text{loc}}^{(m)}$  denote the threshold associated with client  $m$ . The confidence ratio is defined as

$$\gamma_{\text{loc}}^{(m)} = \frac{\text{GLR}_m(b_m, c_m)}{\beta_{\text{loc}}^{(m)}}.$$

The interpretation is straightforward:

- If  $\gamma_{\text{loc}}^{(m)} < 1$ , there is insufficient evidence to certify the local best arm.
- If  $\gamma_{\text{loc}}^{(m)} \geq 1$ , the leader satisfies the local stopping criterion.

Thus, the quantity  $\gamma_{\text{loc}}^{(m)}$  measures how close client  $m$  is to confidently identifying its local best arm.

The values of  $\gamma_{\text{loc}}^{(m)}$  also play an important role in the sampling strategy. Clients whose local confidence ratios are small receive higher priority, since they correspond to the current bottlenecks in the identification process.

## 4.4 Global Best Arm Identification

In addition to identifying the best arm at each client, the server is responsible for identifying the globally optimal arm. Recall that the global mean of arm  $a$  is defined as

$$\mu_a = \frac{1}{M} \sum_{m=1}^M \mu_{a,m},$$

and the global best arm is

$$a^* = \arg \max_{a \in [K]} \mu_a.$$

Since the server does not have access to the raw rewards, all decisions must be made using the privatized statistics communicated by the clients. Let  $\tilde{\mu}_{a,m}$  denote the most recent privatized estimate received from client  $m$ . The server constructs the aggregate estimate

$$\tilde{\mu}_a = \frac{1}{M} \sum_{m=1}^M \tilde{\mu}_{a,m}.$$

Similarly, the effective number of samples associated with arm  $a$  is defined as

$$\tilde{N}_a = \sum_{m=1}^M \tilde{N}_{a,m}.$$

The server maintains a global leader and a global challenger in a manner analogous to the local identification problem.

The global leader is chosen according to the privacy-adjusted upper confidence index

$$B = \arg \max_{a \in [K]} \left( \tilde{\mu}_a + \sqrt{\frac{k_a}{\tilde{N}_a}} + \frac{k_a}{\varepsilon \tilde{N}_a} \right),$$

where

$$k_a = \log_2(\tilde{N}_a).$$

The global challenger is selected as

$$C = \arg \min_{a \neq B} \text{GLR}_{\text{glob}}(B, a).$$

Consequently, the pair  $(B, C)$  represents the most informative global comparison at the current stage of the algorithm.

## 4.5 Sampling Rule

The sampling strategy combines local and global objectives. At any given time, a client may either require additional samples to distinguish its own local leader from

its local challenger or may be requested to assist the server in resolving uncertainty regarding the global best arm.

To determine which objective should be prioritized, each client computes its local confidence ratio

$$\gamma_{\text{loc}}^{(m)} = \frac{\text{GLR}_m(b_m, c_m)}{\beta_{\text{loc}}},$$

while the server computes the global confidence ratio

$$\gamma_{\text{glob}} = \frac{\text{GLR}_{\text{glob}}(B, C)}{\beta_{\text{glob}}}.$$

These quantities measure the relative progress of the local and global identification tasks.

If

$$\gamma_{\text{loc}}^{(m)} < \gamma_{\text{glob}},$$

then the local identification problem is currently the bottleneck. In this case, client  $m$  focuses on distinguishing the local leader from the local challenger and samples from the set

$$\{b_m, c_m\}.$$

Otherwise,

$$\gamma_{\text{loc}}^{(m)} \geq \gamma_{\text{glob}},$$

and the client allocates samples to support the global identification problem. In this case, sampling is restricted to arms that contribute to the comparison between the global leader  $B$  and the global challenger  $C$ .

This adaptive allocation strategy ensures that samples are directed toward whichever component of the identification problem currently lacks sufficient evidence. As a consequence, the algorithm avoids wasting samples on comparisons that have already been resolved with high confidence.

## 4.6 Stopping Rule

The stopping rule determines the earliest time at which sufficient statistical evidence has been accumulated to confidently identify both the local and global best arms.

The construction follows the Track-and-Stop principle of Garivier and Kaufmann. At every stage, generalized likelihood ratio (GLR) statistics are used to quantify the evidence against alternative hypotheses. The algorithm terminates once the corresponding GLR statistics exceed suitably chosen confidence thresholds.

### 4.6.1 Threshold Function

Following [3], we employ a confidence threshold that guarantees  $\delta$ -correctness uniformly over time. Let  $s > 1$  and let  $\zeta(\cdot)$  denote the Riemann zeta function. Define

$$C_G(x) = x + \ln(x).$$

For integers  $n, m \geq 1$ , define the non-private threshold

$$c_k(n, m, \delta) = 2C_G\left(\log\left(\frac{(K-1)\zeta(s)k^s}{\delta}\right)\right) + 2\log(4 + \log n) + 2\log(4 + \log m).$$

The privacy-adjusted threshold is given by

$$c_{\varepsilon, k_1, k_2}(n, m, \delta) = 2c_{k_1 k_2}(n, m, \delta/2) + \frac{1}{n\varepsilon^2} \log\left(\frac{2Kk_1^s \zeta(s)}{\delta}\right)^2 + \frac{1}{m\varepsilon^2} \log\left(\frac{2Kk_2^s \zeta(s)}{\delta}\right)^2.$$

As observed in [3], the threshold admits the approximation

$$c_{\varepsilon, k_1, k_2}(n, m, \delta) \approx 2\log\left(\frac{1}{\delta}\right) + \left(\frac{1}{n} + \frac{1}{m}\right) \frac{\log^2(1/\delta)}{\varepsilon^2}.$$

The first term corresponds to the classical non-private confidence requirement, while the second term captures the additional statistical cost imposed by differential privacy.

### 4.6.2 Local Stopping Rule

For every client  $m$ , let

$$b_m = \arg \max_{a \in [K]} \tilde{\mu}_{a,m}$$

denote the current local leader and let

$$c_m = \arg \min_{a \neq b_m} \text{GLR}_m(b_m, a)$$

denote its challenger.

The local GLR statistic is

$$\text{GLR}_m(b_m, c_m) = \tilde{N}_{b_m,m} d(\tilde{\mu}_{b_m,m}, c) + \tilde{N}_{c_m,m} d(\tilde{\mu}_{c_m,m}, c),$$

where

$$c = \frac{\tilde{N}_{b_m,m} \tilde{\mu}_{b_m,m} + \tilde{N}_{c_m,m} \tilde{\mu}_{c_m,m}}{\tilde{N}_{b_m,m} + \tilde{N}_{c_m,m}}$$

is the pooled empirical mean.

The client declares its local best arm once

$$\text{GLR}_m(b_m, c_m) \geq \beta_{\text{loc}},$$

where

$$\beta_{\text{loc}} = c_{\varepsilon, k_1, k_2} \left( \tilde{N}_{b_m,m}, \tilde{N}_{c_m,m}, \delta \right).$$

### 4.6.3 Global Stopping Rule

Let  $B$  denote the current empirical global leader and let  $C$  denote its most competitive challenger. The server computes the global GLR statistic

$$\text{GLR}_{\text{glob}}(B, C).$$

The global decision is declared whenever

$$\text{GLR}_{\text{glob}}(B, C) \geq \beta_{\text{glob}},$$

where

$$\beta_{\text{glob}} = c_{\varepsilon, k_1, k_2}(N_B, N_C, \delta).$$

The algorithm terminates only when both the local and global stopping conditions are satisfied, namely

$$\text{GLR}_m(b_m, c_m) \geq \beta_{\text{loc}} \quad \forall m \in [M],$$

and

$$\text{GLR}_{\text{glob}}(B, C) \geq \beta_{\text{glob}}.$$

At this point the algorithm outputs

$$\hat{\mathbf{A}} = (\hat{b}_1, \dots, \hat{b}_M, \hat{a}^*).$$

By construction of the threshold function, the probability that any local or global recommendation is incorrect is controlled uniformly over time, yielding a  $\delta$ -correct algorithm.

## 4.7 Complete Algorithm

We now present the complete federated differentially private top-two sampling algorithm. The algorithm operates through a collection of clients and a central server. Clients maintain local estimates and communicate privatized statistics only at the end of doubling episodes. The server aggregates these statistics to construct global estimates and coordinate exploration.

The overall procedure is described through two interacting algorithms. Algorithm 1 describes the operations performed by each client, while Algorithm 2 describes the operations performed at the server.

---

**Algorithm 1** Client  $m$

---

**Require:** Privacy parameter  $\varepsilon$ , confidence level  $\delta$ , number of clients  $M$ , number of arms  $K$

**Ensure:** Local recommendation  $\hat{b}_m$  and communication messages sent to server

- 1: Initialize pull counts  $\tilde{N}_{a,m} = 0$  for all  $a \in [K]$
- 2: Pull each arm once
- 3: **while** not stopped **do**
- 4:     Receive  $(B, C, \gamma_{\text{glob}})$  from server
- 5:     Compute local leader

$$b_m = \arg \max_{a \in [K]} \left( \tilde{\mu}_{a,m} + \sqrt{\frac{k_{a,m}}{\tilde{N}_{a,m}}} + \frac{k_{a,m}}{\varepsilon \tilde{N}_{a,m}} \right)$$

- 6:     Compute local challenger

$$c_m = \arg \min_{a \neq b_m} \text{GLR}_m(b_m, a)$$

- 7:     Compute local confidence ratio

$$\gamma_{\text{loc}}^{(m)} = \frac{\text{GLR}_m(b_m, c_m)}{\beta_{\text{loc}}^{(m)}}$$

- 8:     **if**  $\gamma_{\text{loc}}^{(m)} < \gamma_{\text{glob}}$  **then**
- 9:         Sample uniformly from  $\{b_m, c_m\}$
- 10:    **else**
- 11:         Sample `select_global`( $B, C$ )
- 12:    **end if**
- 13:    Observe reward and update statistics
- 14:    **if** episode for selected arm ends **then**
- 15:         Compute empirical mean  $\hat{\mu}_{a,m}$
- 16:         Release

$$\tilde{\mu}_{a,m} = \hat{\mu}_{a,m} + \text{Lap} \left( \frac{1}{n\varepsilon} \right)$$

- 17:         Forget episode data
- 18:         Send to server

$$(\tilde{\mu}_{a,m}, \tilde{N}_{a,m}, \gamma_{\text{loc}}^{(m)})$$

- 19:     **end if**
  - 20: **end while**
-

---

**Algorithm 2** Server

---

**Require:** Privacy parameter  $\varepsilon$ , confidence level  $\delta$ , number of clients  $M$ , number of arms  $K$

**Ensure:** Global recommendation  $\hat{a}^*$

- 1: Receive initial privatized estimates from all clients
- 2: **while** not stopped **do**
- 3:     Aggregate privatized means

$$\tilde{\mu}_a = \frac{1}{M} \sum_{m=1}^M \tilde{\mu}_{a,m}$$

- 4:     Aggregate pull counts

$$\tilde{N}_a = \frac{1}{M} \sum_{m=1}^M \tilde{N}_{a,m}$$

- 5:     Compute global leader

$$B = \arg \max_{a \in [K]} \left( \tilde{\mu}_a + \sqrt{\frac{k_a}{\tilde{N}_a}} + \frac{k_a}{\varepsilon \tilde{N}_a} \right)$$

- 6:     Compute global challenger

$$C = \arg \min_{a \neq B} \text{GLR}_{\text{glob}}(B, a)$$

- 7:     Compute global confidence ratio

$$\gamma_{\text{glob}} = \frac{\text{GLR}_{\text{glob}}(B, C)}{\beta_{\text{glob}}}$$

- 8:     Broadcast  $(B, C, \gamma_{\text{glob}})$
- 9:     **if**  $\gamma_{\text{glob}} \geq 1$  and  $\gamma_{\text{loc}}^{(m)} \geq 1$  for all  $m$  **then**
- 10:         Stop
- 11:         Output

$$(\hat{b}_1, \dots, \hat{b}_M, \hat{a}^*)$$

- 12:     **end if**
  - 13: **end while**
-



## 4.8 Privacy analysis

**Theorem 4.1** (Privacy guarantee of Fed-AdaP-TT). *Assume that rewards are bounded in  $[0, 1]$ . Then the proposed FED-ADAP-TT algorithm satisfies  $\varepsilon$ -global differential privacy with respect to the neighbouring relation introduced in Definition 2.4.*

*More precisely, for any pair of neighbouring reward tables  $R^T$  and  $\bar{R}^T$ , and for any possible transcript*

$$O = \left( A^T, \hat{a}^\star, \hat{b}_1, \dots, \hat{b}_M, T \right),$$

*we have*

$$\Pi(O \mid R^T) \leq e^\varepsilon \Pi(O \mid \bar{R}^T).$$

*Proof.* Let

$$Z_{1:R} = (Z_1, \dots, Z_R)$$

denote the sequence of all privatized episode means released by the clients during the execution of the algorithm.

The proof consists of three observations.

**Step 1: Every decision is a function of privatized statistics.**

The server never observes raw rewards. All arm-selection decisions, leader and challenger computations, GLR statistics, stopping decisions, and final recommendations are computed exclusively from the privatized episode means exchanged between the clients and the server.

Consequently, there exists a randomized mapping  $F$  such that

$$O = F(Z_{1:R}, \Phi),$$

where  $\Phi$  denotes the internal randomness of the algorithm. Therefore the complete transcript is a post-processing of the privatized release sequence  $Z_{1:R}$ .

**Step 2: Each release is  $\varepsilon$ -DP.**

Consider a particular episode release  $Z_r$ . Suppose that the episode contains the set of rewards

$$S_r = \{R_i : i \in I_r\}.$$

The released statistic is

$$Z_r = \frac{1}{|S_r|} \sum_{i \in I_r} R_i + \eta_r,$$

where

$$\eta_r \sim \text{Lap}\left(\frac{1}{|S_r|\varepsilon}\right).$$

Since rewards lie in  $[0, 1]$ , changing a single reward changes the episode mean by at most

$$\Delta_r = \frac{1}{|S_r|}.$$

Thus the Laplace mechanism guarantees that each release  $Z_r$  is  $\varepsilon$ -differentially private.

**Step 3: Parallel composition.**

The forgetting property ensures that every observed reward contributes to exactly one episode mean. Hence the collections of rewards used by different releases are disjoint.

Furthermore, episode boundaries are determined solely by pull counts and previously released privatized quantities. Therefore the identity of the episode being released is completely determined by the privatized history and does not depend on unobserved raw rewards.

Consider two neighbouring reward tables that differ in exactly one reward value. Because the episode sets are disjoint, the modified reward can affect at most one release, say  $Z_{r^*}$ . All remaining releases are identical under the two neighbouring reward tables.

Therefore,

$$\frac{\Pr(Z_{1:R} \mid R^T)}{\Pr(Z_{1:R} \mid \bar{R}^T)} = \frac{\Pr(Z_{r^*} \mid R^T)}{\Pr(Z_{r^*} \mid \bar{R}^T)} \leq e^\varepsilon,$$

where the final inequality follows from the privacy guarantee of the Laplace mechanism.

Hence the entire release sequence  $Z_{1:R}$  is  $\varepsilon$ -differentially private.

Finally, since the transcript  $O$  is a post-processing of  $Z_{1:R}$ , the post-processing property of differential privacy yields

$$\Pi(O \mid R^T) \leq e^\varepsilon \Pi(O \mid \bar{R}^T).$$

Therefore the proposed algorithm satisfies  $\varepsilon$ -global differential privacy.

□

# Chapter 5

## Experimental Results

This chapter presents the experimental evaluation of the proposed FED-ADAP-TT algorithm. The primary objective of these experiments is to validate the theoretical predictions developed in the previous chapters and to study the practical behaviour of the algorithm under different privacy and problem regimes.

### 5.1 Sample Complexity versus Privacy Budget

The first experiment studies the dependence of the stopping time on the privacy parameter  $\varepsilon$ . Recall that the lower bound derived in Chapter 3 takes the form

$$\mathcal{K}^*(\nu, \varepsilon) = \sup_{\omega} \min \{6\varepsilon T(\omega), D(\omega)\},$$

which implies the asymptotic sample complexity

$$\mathbb{E}[\tau] = \Omega\left(\frac{\log(1/\delta)}{\mathcal{K}^*(\nu, \varepsilon)}\right).$$

The structure of the lower bound immediately suggests two qualitatively different privacy regimes. For sufficiently small values of  $\varepsilon$ , the total variation term  $6\varepsilon T(\omega)$  becomes the bottleneck and therefore

$$\mathbb{E}[\tau] = \Theta\left(\frac{1}{\varepsilon}\right).$$

On the other hand, once  $\varepsilon$  exceeds a critical threshold, the KL-divergence term becomes dominant and the sample complexity becomes essentially independent of

$\varepsilon$ . In this regime, the behaviour of the algorithm approaches that of a non-private best-arm identification procedure.

To verify this prediction, we consider two different problem instances. The first instance is chosen so that the dominant source of difficulty arises from identifying the local best arms. The mean matrix is given by

$$\mu = \begin{bmatrix} 0.40 & 0.45 & 0.50 & 0.58 & 0.63 & 0.72 \\ 0.60 & 0.54 & 0.43 & 0.52 & 0.36 & 0.46 \\ 0.72 & 0.46 & 0.43 & 0.41 & 0.40 & 0.45 \\ 0.80 & 0.52 & 0.56 & 0.48 & 0.53 & 0.54 \\ 0.48 & 0.50 & 0.52 & 0.68 & 0.58 & 0.55 \\ 0.80 & 0.48 & 0.52 & 0.42 & 0.90 & 0.35 \end{bmatrix}.$$

For this instance, the global best arm is relatively well separated, whereas several clients contain local competitors with small gaps. Consequently, local identification constitutes the principal statistical difficulty.

The second instance is constructed so that the global identification problem becomes the dominant challenge. Its mean matrix is

$$\mu = \begin{bmatrix} 0.90 & 0.30 & 0.15 & 0.25 \\ 0.95 & 0.35 & 0.20 & 0.10 \\ 0.35 & 0.65 & 0.20 & 0.10 \\ 0.40 & 0.80 & 0.20 & 0.15 \end{bmatrix}.$$

In this case, the local best arms are relatively easy to identify, whereas distinguishing the globally optimal arm requires aggregating evidence across multiple clients. The global BAI constraint therefore becomes the binding event.

The experimental results are shown in Figure 5.1. In both instances, the behaviour predicted by the lower bound is clearly visible. For small values of  $\varepsilon$ , the stopping time increases rapidly as the privacy constraint becomes more stringent. When represented on the logarithmic scale used in the figure, this region exhibits the characteristic behaviour associated with the theoretical  $1/\varepsilon$  scaling.

As  $\varepsilon$  increases, a transition occurs between the privacy-limited and information-limited regimes. Beyond this transition point, further increases in the privacy budget produce only marginal improvements in sample complexity. The stopping time stabilizes and approaches a constant value, indicating that the privacy constraint is no longer the limiting factor. In this region, the algorithm behaves similarly to its non-private counterpart.

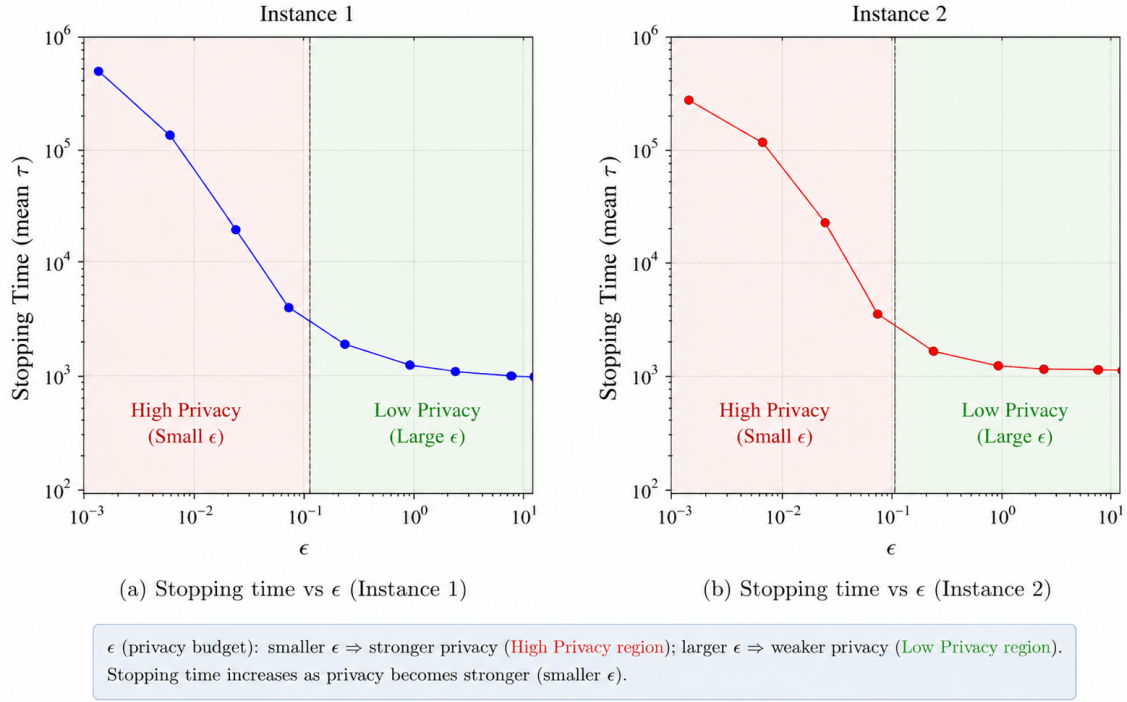


Figure 5.1: Average stopping time as a function of the privacy parameter  $\epsilon$  for two different problem instances. The left panel corresponds to an instance in which local best-arm identification is the dominant difficulty, while the right panel corresponds to an instance in which the global best-arm identification problem is the dominant bottleneck.

An important observation is that the same qualitative behaviour is present in both instances despite the fact that different events govern their complexity. In the first instance, the dominant contribution to the lower bound arises from local best-arm identification events, whereas in the second instance it arises from the global best-arm identification event. The appearance of the same two-regime structure in both settings confirms that the privacy-information trade-off predicted by the lower bound is intrinsic to the problem formulation and is not an artifact of a particular instance geometry.

## 5.2 Effect of Federation

A central motivation for federated best-arm identification is the ability to collect information simultaneously from multiple clients while preserving the privacy of their

local observations. Unlike the classical single-agent setting, a federated system benefits from parallel data acquisition across clients, potentially reducing the amount of time required to accumulate sufficient statistical evidence. At the same time, increasing the number of clients introduces additional local identification tasks and increases the communication burden between the clients and the server. Understanding the net effect of federation is therefore an important empirical question.

To investigate this phenomenon, we study the performance of FED-ADAP-TT as the number of clients increases while keeping both the confidence parameter and privacy budget fixed at  $\delta = \epsilon = 0.05$ .

Three problem instances are considered, corresponding to  $M = 6$ ,  $M = 12$ , and  $M = 18$  clients respectively. The latter two instances are obtained by augmenting the original six-client instance with additional clients exhibiting heterogeneous local preferences. This construction preserves the federated nature of the problem while allowing us to examine the effect of increasing the scale of the system.

Table 5.1 reports the theoretical characteristic time  $T^*$ , the empirical stopping time, and the average number of communication rounds required by the algorithm.

| Clients | $T^*$              | $\tau$ (mean $\pm$ std)                 | Comm. Rounds (mean $\pm$ std) |
|---------|--------------------|-----------------------------------------|-------------------------------|
| 6       | $7.37 \times 10^3$ | $5.81 \times 10^5 \pm 1.63 \times 10^5$ | $453.6 \pm 5.37$              |
| 12      | $1.69 \times 10^4$ | $5.88 \times 10^5 \pm 1.39 \times 10^5$ | $860.6 \pm 12.17$             |
| 18      | $2.84 \times 10^4$ | $7.96 \times 10^5 \pm 2.59 \times 10^5$ | $1244.58 \pm 10.63$           |

Table 5.1: Performance as the number of clients increases.

The results reveal several key insights. First, the characteristic time  $T^*$  grows alongside the client count. This makes sense because each new client adds a new local best-arm identification problem that must be resolved before termination, making the underlying optimization problem more constrained and complex.

The actual empirical stopping time shows a more interesting trend. Moving from six to twelve clients, the average stopping time barely changes, even though  $T^*$  increases significantly. This happens because parallel sampling provides extra data that balances out the higher statistical difficulty. Because all clients gather rewards at the same time, the server gets more data during each communication round, meaning the benefits of parallelism largely offset the added complexity.

However, once the client count reaches eighteen, the stopping time starts to climb. At this scale, managing a larger set of local identification problems becomes costly, and the influx of new data can no longer completely cancel out the added difficulty.

Even so, the sample complexity grows much slower than the theoretical characteristic time, showing that federation still offers a clear advantage.

This impact is especially clear when looking at communication rounds. As anticipated, communication costs scale roughly linearly with the number of clients. More clients mean more local decisions, more challenger updates, and more data moving back and forth before reaching a confident global decision. This underscores a core trade-off in federated best-arm identification: adding clients boosts data collection through parallelism but raises communication overhead.

In short, these results prove that federation improves data collection efficiency by leveraging parallel operations across clients. While more clients increase the problem’s inherent complexity, the simultaneous data streams mitigate most of this burden. The experiments validate the core idea of federated best-arm identification—that distributed data gathering delivers strong statistical advantages while preserving privacy.

### 5.3 Effect of the Confidence Parameter

In this experiment, we fix the privacy budget at  $\varepsilon = 0.005$  and consider the six-client Bernoulli instance

$$\begin{bmatrix} 0.40 & 0.45 & 0.50 & 0.58 & 0.63 & 0.72 \\ 0.60 & 0.54 & 0.43 & 0.52 & 0.36 & 0.46 \\ 0.72 & 0.46 & 0.43 & 0.41 & 0.40 & 0.45 \\ 0.80 & 0.52 & 0.56 & 0.48 & 0.53 & 0.54 \\ 0.48 & 0.50 & 0.52 & 0.68 & 0.58 & 0.55 \\ 0.80 & 0.48 & 0.52 & 0.42 & 0.90 & 0.35 \end{bmatrix}.$$

The objective is to study how the stopping time varies with the confidence level. Rather than reporting  $\delta$  directly, we use the quantity  $\log(1/\delta)$ , which appears in the lower bound

$$\mathbb{E}_\nu[\tau] \gtrsim \frac{\log(1/\delta)}{\mathcal{K}^\star(\nu, \varepsilon)}.$$

The privacy parameter and problem instance are kept fixed while  $\log(1/\delta)$  is varied from 1 to 10.

Table 5.2 shows that both the stopping time and the communication complexity increase as the confidence requirement becomes more stringent. As predicted by



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| $\log(1/\delta)$ | $T^*$              | $\tau$ (mean $\pm$ std)       | Comm. Rounds (mean $\pm$ std) |
|------------------|--------------------|-------------------------------|-------------------------------|
| 1                | $7.37 \times 10^3$ | $(8.06 \pm 1.47) \times 10^4$ | $471.7 \pm 5.4$               |
| 2                | $1.47 \times 10^4$ | $(8.59 \pm 1.43) \times 10^4$ | $477.0 \pm 5.8$               |
| 3                | $2.21 \times 10^4$ | $(9.63 \pm 0.86) \times 10^4$ | $480.9 \pm 6.2$               |
| 4                | $2.95 \times 10^4$ | $(9.91 \pm 0.77) \times 10^4$ | $483.4 \pm 6.8$               |
| 5                | $3.68 \times 10^4$ | $(9.96 \pm 0.70) \times 10^4$ | $485.7 \pm 7.0$               |
| 6                | $4.42 \times 10^4$ | $(10.5 \pm 1.07) \times 10^4$ | $487.5 \pm 6.8$               |
| 7                | $5.16 \times 10^4$ | $(11.5 \pm 0.90) \times 10^4$ | $489.6 \pm 6.8$               |
| 8                | $5.89 \times 10^4$ | $(12.1 \pm 0.61) \times 10^4$ | $491.6 \pm 7.5$               |
| 9                | $6.63 \times 10^4$ | $(12.5 \pm 0.86) \times 10^4$ | $494.3 \pm 7.8$               |
| 10               | $7.37 \times 10^4$ | $(12.8 \pm 0.99) \times 10^4$ | $497.2 \pm 8.0$               |

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Table 5.2: Effect of the confidence parameter on stopping time and communication complexity for  $\varepsilon = 0.005$ . Here the first column reports  $\log(1/\delta)$ .

the lower-bound analysis, the dominant scaling is approximately linear in  $\log(1/\delta)$ . Increasing the confidence level requires the algorithm to accumulate more statistical evidence before termination, resulting in a larger number of arm pulls and communication rounds. The results therefore corroborate the asymptotic prediction that the sample complexity grows proportionally to  $\log(1/\delta)$  while the dependence on the problem instance is captured through the characteristic constant  $\mathcal{K}^*(\nu, \varepsilon)$ .

## 5.4 Conclusion

This thesis studied the problem of fixed-confidence best arm identification in a federated multi-armed bandit setting under differential privacy constraints. Motivated by applications where data is naturally distributed across multiple agents and direct sharing of observations is undesirable, we considered the task of identifying both the local best arm at each client and the globally optimal arm while preserving privacy.

We began by reviewing the classical best arm identification problem and the fundamental lower bounds that govern its sample complexity. In particular, we discussed the game-theoretic characterization of the characteristic time and the algorithmic principles underlying Track-and-Stop and Top-Two sampling strategies. These ideas served as the foundation for the federated formulation considered in this work.

Next, we introduced the federated best arm identification problem and formalized the concept of  $\varepsilon$ -global differential privacy. By combining the change-of-measure arguments of Garivier and Kaufmann with the privacy-sensitive transportation inequalities from Azize et al., we established an instance-dependent lower bound for this private federated environment. This characterization explicitly highlights the

trade-off between statistical information (via Kullback–Leibler divergence) and privacy (via total variation distance). For Bernoulli bandits, we simplified this lower bound and analyzed its structural properties, demonstrating that the optimization problem retains a concave saddle-point structure. We also identified two distinct regimes: a privacy-limited regime where sample complexity scales as  $1/\varepsilon$ , and a low-privacy regime that recovers classical, non-private complexity.

Based on these insights, we developed a federated differentially private best arm identification algorithm that combines a Top-Two sampling principle with Laplace perturbations. The algorithm uses separate local and global generalized likelihood ratio statistics alongside privacy-adjusted stopping thresholds. Using a composition argument based on the Laplace mechanism, we verified that this framework satisfies the required  $\varepsilon$ -global differential privacy constraint.

Our empirical findings back up these theoretical predictions. The experiments confirm the existence of the two predicted privacy regimes, validate the expected dependence on the confidence parameter  $\delta$ , and show how federation impacts both sample complexity and communication needs. The observed behavior aligns well with the qualitative predictions of the lower-bound analysis across various problem instances.

Several key questions remain for future research. The most immediate step is to provide a rigorous proof that the proposed algorithm is  $\delta$ -PAC. This requires establishing concentration guarantees for the privatized estimates and proving that the stopping rule limits the probability of incorrect identification uniformly over time. We also need a complete achievability analysis; specifically, it remains to be seen whether this algorithm reaches the lower bound asymptotically, or if a Track-and-Stop strategy can be adapted to the federated private setting to achieve asymptotic optimality.

Another direction is designing adaptive privacy mechanisms that allocate the privacy budget dynamically across clients and communication rounds. Future work could also extend this analysis from Bernoulli rewards to broader exponential families, investigate communication-efficient variants, and derive finite-time performance guarantees.

In short, this study offers an initial look at how exploration, federation, and privacy interact in best arm identification. The lower-bound characterization, algorithmic setup, and experimental results established here offer a clear foundation for

developing statistically efficient, privacy-preserving systems for sequential decision-making.

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